Epidemiological Section.

April, 1908.

On an Epidemic of Small-pox of Irregular Type in Trinidad during 1902–4.

By R. SEHEULT, M.B.

SMALL-POX was so prevalent in prevaccination times that hardly anyone escaped the disease. It entered the palace of the king with the same freedom as it did the hovel of the peasant; it penetrated everywhere, carrying desolation with it. Those who escaped death were left disfigured or crippled for life; almost every face was seamed and scarred, and on every side were met the blinded victims of the scourge. At times whole towns were depopulated. When the contagion fell upon virgin soil it raged with special virulence and wrought dreadful havoc. Among the black races, whole tribes were extirpated; its ravages were then fearful to contemplate, and the mortality which followed in its train was appalling. Macaulay, in his "History of England," thus alludes to this scourge in speaking of Queen Mary's death from it in 1694 :—

> That disease, over which science has achieved a succession of glorious and beneficent victories, was then the most terrible of all the ministers of death. The havoc of the plague had been more rapid, but the plague had visited our shores only once or twice within living memory. The small-pox was always present, filling the churchyard with corpses, tormenting with constant fears all those whom it had not yet stricken, leaving on those whose lives it spared the hideous traces of its power, turning the babe into a changeling at which the mother shuddered, and making the eyes and cheeks of the betrothed maiden objects of horror to her lover.

Even more impresssive than this classical picture of the great historian is the evidence presented by statistics, in which is crystallized the experience of entire nations. The features of this loathsome and destructive disease were then familiar even to the man in the street, and medical men had ample opportunities of becoming thoroughly acquainted with its various manifestations; but since the discovery, or, more correctly speaking, the introduction, of vaccination by the immortal Jenner, this most dreaded of all the infectious diseases has been by degrees stamped out in all civilized countries, or at any rate its prevalence has been lessened to such an extent that there are nowadays many experienced physicians who have never seen a case. Furthermore, the practice of vaccination has rendered the diagnosis more difficult, as the phases of the disease have been made by it far more numerous and intricate than they were before. It is not surprising, therefore, since its epidemic character has been so greatly modified by vaccination and other causes, that difficulties in recognizing its true nature are experienced at times.

After the great pandemic of 1871—2, small-pox did not again appear in the Colony of Trinidad until April, 1902, when the disease was introduced in a very mild and irregular form, giving rise to considerable diversity of opinion in regard to its nature. Among the sixty medical practitioners in the island there were not eight who had had any experience of this disorder; and even the doyen of the medical faculty there, who had witnessed the terrible ravages of small-pox in 1871—2, was misled by the aberrant symptomatology of the disease in this latter epidemic, and failed to recognize its true nature. The outbreak, which commenced in 1902, appears especially deserving of detailed study, having in view the interesting points connected with the origin and spread of the epidemic, the somewhat anomalous features presented by the disease, the instructive results obtained in regard to the relation of vaccination to it, and, above all, the mortality, which was so strikingly low.

The following account of the origin of the epidemic in Trinidad is taken from a pamphlet by Dr. Dickson, the Assistant Medical Officer of Health, and Dr. Lassalle, Assistant Surgeon, Colonial Hospital, Port-of-Spain, entitled "Varioloid Varicella in Trinidad." This paper was read at the meeting of the British Medical Association at Swansea in 1903:—

The first case of which there is a record was that of an inmate of the Lunatic Asylum, St. Ann's. The asylum is situated in an isolated position, beyond the limits of the town. This patient had been an inmate of the asylum for some years, and developed the disease on April 16, 1902. The case was isolated on the appearance of the rash, but other cases appeared during May, June, July and August, until nineteen inmates and attendants, all adults, were affected. The source of infection could not be traced, and must have been either a visitor or attendant who had a mild attack and escaped notice. The cases were returned as "varicella," but the medical superintendent has since reported that they were similar to the cases of eruptive fever now occurring, and in one instance—that of an attendant who had the disease in August, 1902—a few pigmented marks identical in appearance with the macules already described were visible up to a month ago. It is of interest, that of the nineteen cases, ten were in vaccinated and six in unvaccinated persons, and in three the evidence of vaccination was doubtful. The patients most severely attacked were an inmate vaccinated in infancy and an attendant revaccinated in 1898, and showing three good marks of successful vaccination.[1]

On May 2, 1902, a similar case in an adult was reported from Woodbrook, a suburb to the west of Port-of-Spain. Cases next occurred in Dundonald Street, in the north-west of the town, in September. Early in October a woman who lived in a barrack-yard in the south-east of the town developed the disease within a fortnight of her arrival from Yrapa, in Venezuela. About the third week of October a case, that of a trader who had recently come from Guiria, in Venezuela, occurred in Duke Street, about the middle of the town. Both of these cases lived in densely tenanted barrack-yards, did not seek medical aid, and were not reported at the time; other inmates of these yards were subsequently affected, but this fact was discovered only in the early part of December and after they had recovered. During November eleven cases occurred in the middle and south-east of the town, and though in all probability the two cases above quoted were the sources of infection, yet there is ground for believing that in three of the cases the contagion was derived from other sources. Five of the cases occurred in one yard in Duncan Street in the first week of November, and of these, two who showed the most distinct vaccination marks were most severely attacked. During December eight cases were reported from the eastern, south-eastern and middle portions of the town. Of these, one was a vagrant who developed the disease within a week of his arrival from Yrapa. In January, 1903, a house to house inspection was instituted, other cases were discovered in various parts of the town, and the disease began to assume epidemic proportions. At first the majority of persons affected were hucksters, sailors and quay labourers, that is, were of that class of the population which would earliest be exposed to contact with an imported disease.

Reports of the prevalence of a similar disease in the adjoining coastal villages of Venezuela had for some time been circulated, and early in February information was received that several deaths had occurred, and that the disease was now stated to be variola. With the view of obtaining accurate information a commission of two medical practitioners, one of whom had had extensive experience of small-pox, was sent to Venezuela to investigate and report upon the nature of the eruptive

[1] With regard to the influence of vaccination *see* p. 281.

fever prevalent there. The commissioners visited Yrapa and Guiria. The following extracts are taken from their report :—

The disease had existed in Yrapa for nearly a year and had not varied in character, that is, had always been a mild affection. Two deaths had occurred in the country around; one was that of a chronic alcoholic, the other died probably more from privation and neglect than anything else. We visited Messrs. Fournelli and Cottin, both Frenchmen long established in Venezuela. Their household had not been attacked, and they were under the impression that Europeans were spared. Mr. Fournelli stated that in Carupano, where there was a large European community and where this disease had been very prevalent, no European had been attacked. These gentlemen informed us that no alarm was ever felt at Yrapa about the sickness, that they called it "lechina" (Spanish for chicken-pox). As a proof of the mildness of the disease they referred to the attack of Guiria by the revolutionary forces from Yrapa, where many of the troopers, though covered with the eruption, carried their Mausers cheerfully to battle.

The commissioners came to the conclusion that the disease was exactly of the same nature as that occurring in Port-of-Spain, and that it was not small-pox. They expressed the opinion that the disease was imported into Trinidad from Yrapa. There is a large daily passenger and trade traffic between Port-of-Spain and the villages on the adjoining Venezuelan coast, and the voyage does not occupy more than a day. Under these circumstances, and in view of the instances above quoted, there seems to be little doubt that the disease was introduced into Trinidad from Venezuela. I may here add the population of Yrapa is about 12,000 and is practically unvaccinated.

So aberrant and misleading were the clinical features of the disease that its real nature was unrecognized in Trinidad except by a few. As I had charge of the isolation hospital for seven months, and treated 564 cases, I had the opportunity of studying closely its various manifestations. I was also placed in charge of the maternity ward, where fifty-one women, who had the disease during pregnancy, were delivered.

Three different theories were advanced to explain the nature of the "eruptive fever." At the commencement of the epidemic, and indeed for a considerable time after, many of the medical men in the Colony considered the disease to be chicken-pox of an aggravated form; the coexistence of syphilis and other constitutional taints, as well as the presence of diathetic tendencies, were put forward to explain the unusual severity of many of the cases. This theory was eventually abandoned by all.

Those who could not countenance or accept this view of the nature of the disease, and yet did not feel justified in considering it to be small-pox, suggested the possibility of the existence of a hybrid between variola and varicella, just as rubella was once considered by Schönlein and other writers to be a hybrid between scarlatina and morbilli. This theory, however, was never seriously maintained, but many were of opinion that the disease was a specific entity, and called it "varioloid varicella," owing to its supposed similarity to an eruptive fever which occurred in epidemic form many years ago in Jamaica, and was described under that name by Dr. Izett Anderson, of Kingston.

I need not comment upon the name "varioloid varicella," which is wholly unscientific and misleading, but some reference to the possibility of the existence of a new disease in the form assumed by this epidemic may not be out of place here. It is well known that at one time measles, scarlatina, rubella and the "fourth disease" were included under one name and were regarded as one malady. With the progress of medical science they were gradually differentiated one from another, so that at the present day they are considered to be perfectly distinct and definite diseases.

It was not until the close of the seventeenth century that scarlet fever was distinguished from measles, whilst the differences between these two diseases and rubella were fully indicated only about the middle of the eighteenth century, when that disease became known as "roseola." The existence of the "fourth disease" as a specific entity has been claimed within very recent years.

Similarly small-pox was for a long period confounded with measles, and even in the sixteenth century, when the former disease was generally recognized, errors of diagnosis were not infrequent. English writers in the early part of the eighteenth century mention varicella as a variety of small-pox, but the end of that century saw the differences between them clearly established.

Can the eruptive fever which forms the subject of this paper be regarded as the "third disease" in the second group of infectious diseases which I have mentioned above, taking its place between varicella and variola? I think not.

The evolution of the diagnosis of the infectious fevers was no doubt in the main due to careful clinical observation, but in those instances, where inherent difficulties of diagnosis existed by reason of very close resemblances, the application of Cullen's law was necessary. "One attack of an eruptive fever entails immunity from a second attack in the same individual during childhood." This law has been regarded as a means of differentiating some of the very closely allied eruptive fevers, and, indeed, as the final test in their elucidation; even now, where the bacteriologist fails to enlighten in such cases, it may prove a very valuable test. The opportunity for the application of this principle in the present case has not arisen, but we have in vaccination a somewhat analogous and equally convincing method of differentiation which can be applied to distinguish other eruptive diseases from variola. Vaccinia and variola are mutually protective, and if the same relation exists, as I shall endeavour to prove, between vaccinia and the disease under review, it is reasonable to infer the identity of the latter with variola.

Although some of the characteristics of the Trinidad epidemic were very unusual and aberrant, yet the more salient features of the disorder were identical with those of small-pox, so that apart from the vaccination test there are grounds for the belief that the two diseases differed only in type. In connection with the theory that the disease was a new malady, the recent volcanic disturbances in the West Indies, notably in Martinique and St. Vincent, appealed to the popular mind, and the disorder was promptly attributed to these convulsions of Nature.

The third view on the subject, and in my opinion the correct one, as I have already indicated, was that the prevailing eruptive fever was an irregular form of small-pox. To Dr. Masson is due the honour of having been the first to recognize the variolous nature of the disease. Early in November, 1902, he was called to see, in Duncan Street, Port-of-Spain, a few cases of an eruptive fever which he at once suspected to be small-pox. In this the Acting Surgeon-General of the Colony and the Assistant Medical Officer of Health and others who saw these cases did not concur; they held the opinion that the disease was chicken-pox. In accordance with this official declaration no steps whatever were taken to prevent the spread of the disease or to circumscribe its area of infection until much later, when it became so widespread as to be almost beyond control. On the other hand it must be admitted that it would have been difficult to prevent dissemination in view of the extreme mildness of a large proportion of the cases. In any case the failure on the part of the Health Department to recognize the true nature of the disease led to its wide diffusion in the town and its invasion of the country districts. This was a blessing in disguise, for the disease not only retained its mild character throughout the epidemic but it spread far and wide in the country, so that a large proportion of the population have become immune from small-pox at a very small sacrifice of lives, through protection afforded either by an attack of the disease or by the operation of vaccination, which the people largely availed themselves of. Fortunately the clamour of the anti-vaccinationists has not yet reached this Colony, nor does a "conscience clause" exist in our Vaccination Ordinance. Dr. Masson was not satisfied with the decision arrived at by the health authorities. Early in December, 1902, he visited Barbados, another West Indian island, which was in the throes of a small-pox epidemic of a more or less mild form, with the object of studying its clinical features and for the purpose of comparing them with those of the cases which he had seen in Port-of-Spain. His observations in Barbados only confirmed his former views on the matter. At first the medical men in Trinidad felt great difficulty in accepting his diagnosis owing to the unusual and variant features of the disease, but subsequently, when they became more intimately acquainted with the epidemic, many recognized the correctness of his contention.

In the meantime the disease, which had hitherto spread very slowly—so slowly that it did not attract any particular attention—began to assume epidemic proportions in the town at the beginning of the year 1903, and to cause much alarm and anxiety to the authorities. Early in 1903 certain measures were adopted to repress its growth. As many cases as possible were sent to the Isolation Hospital, but for want of accommodation the vast majority of the patients were treated at their homes. In February, 1903, two medical men were specially appointed for this purpose. Contacts were vaccinated, and revaccination was encouraged generally. The disease continued to spread, and, owing to its mild character, many of those affected by it were seen in the streets in its various stages, and in some instances they were actually able to pursue their daily labours. In order to protect the public health against the so-called "varioloid varicella," certain regulations were made by His Excellency the Governor in Executive Council, but these were never strictly enforced, and the disease followed untrammelled its own course and spread throughout the whole island.

From April 16, 1902, when the first case was discovered, to December 31, 1902, there were only sixty cases reported. The extremely slow and insidious spread of the epidemic was one of the circumstances which led the profession to persist in the error of diagnosis. The negro race is known to be especially susceptible to the contagion of small-pox, and when their conditions of life in crowded barrack-yards and their ignorance of ordinary sanitation are considered, the slow advance of the outbreak is very remarkable. It was only in January that the outbreak began to assume epidemic form, reaching its maximum height in May, and then gradually declining until its entire disappearance from the town in November, 1903, and from the country in January, 1904.

It must be borne in mind that although the native population is a fairly well vaccinated one, owing to the rigidly enforced Vaccination Ordinance, there is a large influx of unvaccinated immigrants from neighbouring islands and from Venezuela, where apparently vaccination is not in great favour. For ten years ending in 1900 the average proportion of vaccinations to births in Trinidad was 83·11 per cent. In the year 1898 the corresponding proportion was 96·48 per cent. Such a result is not probably equalled in any other part of the British dominions. Among the 564 cases which came under my observation in the isolation wards only 118 were Trinidadians, the rest being aliens, and of these 254 hailed from Barbados (*see* Table I.). The protection afforded to the Colony against an epidemic of small-pox will not be complete until provision is made for the successful vaccination of the large number of unvaccinated persons coming from the other Colonies and the adjacent continent. Every immigrant should be required to exhibit proof of successful vaccination before being allowed to land in this country, as is done in some States of America.

The slow spread of the epidemic was due to the slight infectivity of the disease. In many cases the contagion or virus seemed to require intimate contact for its transmission from one person to another, and even then it was remarkable how frequently instances were found in which such contacts escaped infection. The Assistant Medical Officer of Health, in a pamphlet already referred to, mentions that in several instances in barrack-yards persons in close association with those affected by the disease did not contract it and subsequently reacted to vaccination. Such a case came under my own observation. A large number of patients were admitted to the general wards of the Colonial Hospital in the incubation or invasion period and were removed to the Isolation Hospital, only a day, or sometimes two days, after the appearance of the rash, and yet no fresh infection took place in these wards. Two cases which developed the disease in the House of Refuge were transferred to the Isolation Hospital in the vesicular stage and none of the other inmates contracted the disease. Four cases which were sent to the male and female prisons in the incubation period were removed to the small-pox wards only after the rash had appeared, and yet there was no spread of the disease in these institutions, although there was no disinfection of any of these buildings. It was frequently observed that children born of mothers in the invasion period or early eruption stage of the disease escaped infection when they were removed from the mother within two or three days after birth, but when left with her until pustulation or desquamation had commenced they invariably contracted the disease. The isolation wards were only 99 ft. from two of the nearest general wards of the Colonial Hospital, and only one patient in each of these wards developed the disease; one was thirty days and the other sixty-five days in hospital before the initial symptoms of small-pox showed themselves. It may be remarked that there were at this period more than 100 small-pox patients in all stages of the disease under treatment in the isolation wards. From these observations it may be inferred that the infectivity of the disease was slight, that the most active period of infection was during pustulation and desquamation, and also that aerial convection, which is held by some recent observers to play an important part in the dissemination of small-pox, was apparently not concerned as a factor in the diffusion of this epidemic.

The mode of spread of the disease to the country districts was also very interesting. It was only in January that the disease occurred in two districts, Tacarigua and Blanchisseuse, which are widely separated from each other. The cases were derived from Port-of-Spain, and occurred on January 8 in the one and on January 24 in the other. The next cases in these districts occurred on February 5 and 12 respectively, and no further cases appeared until April 3 and May 1. The first case which occurred in Blanchisseuse was that of a man who arrived on January 21 from Port-of-Spain, where he had associated with persons suffering from the disease. Three days after his arrival he developed the initial symptoms of the disease.

In March several other districts became infected; the diffusion of the contagion to the country districts in March is readily accounted for by the fact that there is always a large influx of country visitors to the town to witness the annual "carnival," which is held at this period of the year.

The first case which was admitted to the Isolation Hospital was that of a woman who was received into one of the general wards of the Colonial Hospital with an infant 35 days old, on November 22, 1902. On December 4 she developed the prodromal symptoms of small-pox, and was then transferred to an isolated room with her infant. I afterwards discovered that this patient had come from a house where there were several cases of "eruptive fever." Her child developed the disease on December 21, 1902, that is, seventeen days after the mother had shown symptoms of small-pox. On January 4, 1903, it was found necessary, on account of the number of cases seeking admission to hospital, to provide further accommodation. Accordingly a ward containing sixteen beds was opened on that day, but at the end of February it had become so overcrowded that another with twenty-two beds was provided on February 28. On March 1, forty-two patients were under treatment. Owing to the rapid spread of the epidemic during this month both wards soon became overcrowded. On March 19 there were no less than sixty-six cases in hospital, so that only the urgent cases were then admitted. On March 27 the number had risen to eighty-six. A third ward with seventy-five beds was opened, but in a very short time this increased accommodation was barely adequate, for on April 2 there were no less than 103 cases under treatment. In May a gradual decrease in the number of cases seeking admission began to take place, and this continued until October. The last patient was discharged on the 19th of that month.

This eruptive fever, as already mentioned, was at its onset officially declared to be chicken-pox, but this diagnosis was revised and altered in the month of March, 1902. The disease then became known as "varioloid varicella," a name which it bore to the end of the epidemic. These diagnoses were accepted without demur by almost all the medical men in the island. Early in 1903 disquieting rumours and conflicting views on the subject of the "Trinidad eruptive fever" determined the government of Barbados to send Dr. Bridger, the medical officer in charge of their small-pox hospital, as a special commissioner to investigate and report upon it. He arrived in Port-of-Spain on February 2, and furnished the government of Trinidad before his departure on March 8 with a report in which he declared the disease to be small-pox of a very mild type. Two days after the receipt of this communication, a meeting of the medical board of the Island was convened, at the special request of His Excellency the Governor, for an expression of opinion upon it. Thirty-four members attended the meeting, the report was read, and after a full discussion on the subject the following resolution was passed with only three dissentients:—

"That no such disease as mild small-pox exists in epidemic form, and that the eruptive fever now prevailing in Trinidad is not small-pox."

Such, then, was the almost unanimous view of the medical profession in Trinidad in regard to the epidemic at that period. The difference of opinion between the Barbados Commissioner and the Trinidad medical practitioners gave rise to a bitter controversy; the columns of the Press of both islands became the channel of much abuse and recrimination. Severe comments were made in some of the British medical journals, which reflected little credit on the diagnostic acumen of the West Indian medical practitioners, and much ridicule was levelled at the profession.

In justification, or rather in extenuation, of the doubt and hesitancy as to the nature of the epidemic, it may be stated that anomalous forms of eruptive fevers, and especially of small-pox, have, at all times, presented similar difficulties of diagnosis, even to experienced observers, causing in many instances much diversity of opinion. Further on I shall refer to two epidemics of a peculiar form of small-pox, popularly known as "swine-pox" and "pearl-pox" respectively, which occurred in Jenner's time. We find in the *Proceedings of the Epidemiological Society of London* a paper entitled "Varioloid Varicella in Jamaica," which was read by Dr. Izett Anderson before that Society in 1867. He describes under that name an eruptive fever which occurred in epidemic form in Jamaica in 1863. He states that in some cases the eruption was apparently that of simple varicella, whilst in others the "inexperienced" would have pronounced it to be that of "variola." The disease attacked young and old, the vaccinated as well as the unvaccinated, and even one or two persons who had had small-pox in 1852, that is, eleven years previously. There was no constitutional disturbance in the majority of the cases, and no necessity to confine the patients to bed. Some malaise and feverishness, but no continued fever of any intensity, preceded the rash; the fever existed for two days, and papules appeared on the third day, usually first on the face in the severer cases, and within twenty-four to forty-eight hours they became vesicles, with sometimes a depression in their centre; the vesicles were then transformed into pustules. The full development of the eruption was attained on the fifth or sixth day of the disease and desquamation followed. Macules and pitting sometimes resulted. The larynx, and in one or two instances the conjunctivæ, were occasionally affected. Secondary fever or anything approaching to it was almost always absent. In the mild cases the vesicles aborted. The epidemic lasted four or five months, and was apparently unattended by any mortality. The disease originated in a penitentiary, and no source of infection from outside could be traced; a fortnight after the appearance of the first case a boys' reformatory, three miles away from the penitentiary, with which there was daily communication, became infected, and forty of the inmates contracted the disease. About a fortnight after the first case appeared in the boys' reformatory the disease broke out in the girls' reformatory, which was half a mile away, and thirty of the inmates were attacked. The disease apparently did not spread to any extent among the general public, although there was communication between the infected institutions and the outside world.

It would appear that the disease was not invariably regarded as varioloid varicella, for in a memorandum Dr. Bowerbank, of Kingston, writes in 1863 :—

> We are at present suffering from a severe influenza and also from a most peculiar epidemic of varicella, I suppose. To me it looks much more like "varioloid" or modified small-pox. Most of the vesicles suppurate and in some instances are distinctly umbilicated and are sometimes confluent. I never saw varicella like this before.

In connection with this epidemic in Jamaica it is interesting to note that a fatal form of small-pox, which was introduced from Colon, followed in 1864. It would be interesting to know whether those who were attacked in the previous epidemic were affected by this fatal form of the disease. The eruptive fever described by Dr. Anderson certainly bears very close resemblance to that which broke out here, but it appears to have been milder in type.

Again, more recently, in the *Lancet* of October 22, 1898, Drs. Thomson and Brownlee record their observations on an infectious disorder in Lascars, having close relations with small-pox and chicken-pox. This infectious disorder appeared to resemble both of these diseases in certain respects and yet to possess symptoms alien to both. After careful consideration the diagnoses of small-pox and chicken-pox were excluded and the disease was regarded by these observers as a specific entity.

I do not share with these observers the opinion that the Glasgow epidemic was identical in nature with that which was reported by Dr. Anderson. The differences between it and the Trinidad eruptive fever are even more marked.

In a pamphlet reprinted from the *Journal of the American Medical Association*, August 3, 1901, Dr. Heman Spalding, Chief Medical Inspector, Department of Health, Chicago, discusses the diagnosis of a mild and irregular form of small-pox which broke out in the United States in 1899. The following extract from this paper indicates the difference of opinion which existed in various parts of the United States in regard to the nature of that outbreak :—

> From March 9, 1899, to June, 1901, 310 cases of small-pox have been found in Chicago; sixty-four of these, in various stages of the disease, were imported

into the city from nineteen of the surrounding States, and the cases came from as far east as New York and from as far west as California. In the meantime I visited three of the neighbouring States, where the diagnosis of this disease, variously called *impetigo contagiosa*, "giant chicken-pox," "Cuban itch," or some other indefinite name, was in dispute. With this opportunity of observing cases from such widespread and various sources, I think it is fair to assume that the disease we call small-pox in Chicago is the same disease which has been the subject of controversy in all parts of the United States.

In the *British Medical Journal* of May 11, 1901, Dr. Montizambert, Director-General of the Public Health Department, Ottawa, speaks of a mild type of small-pox which was undoubtedly of the same nature as that referred to by Dr. Spalding, and probably similar to that which broke out in Trinidad. In this article, which is entitled "Notes on a Mild Type of Small-pox (Variola ambulans)," the author writes:—

The Dominion of Canada is now being threatened with, and in some cases invaded by, small-pox from her neighbour, the United States. It began on this continent several years ago in the United States, the southern States especially. It has gradually spread northwards. Its origin is difficult to establish, either as to time or place, with any historical accuracy. It has been attributed by many to soldiers returning from Cuba or from the Philippines. But it is certain that it was prevalent in the United States before the beginning of the war between that country and Spain. The difficulty in tracing back its history is due in great part to the unusual mildness of the type. Many cases were diagnosed as chicken-pox, many as German measles. In many of the lumber camps it went by the name of "cedar itch."

In the *Lancet* of July 4, 1903, p. 65, reference is made to an outbreak of disease in Cambridge which appears to have caused some doubt and uncertainty in the minds of the health authorities. The main features of the disease seemed at first incompatible with small-pox, and the diagnosis of chicken-pox was made; as the epidemic increased in severity, however, expert advice was sought, and Dr. Wanklyn, the referee to the Metropolitan Asylums Board, who was invited to examine the cases, reported the disease to be undoubtedly modified small-pox.

From these few examples it will be seen that sometimes irregular forms of small-pox present great difficulty as regards diagnosis, raising doubt even in the minds of the experienced. The strictures, therefore, which were passed by those who were not confronted by this atypical variety of small-pox were unmerited and unjustified. The nature of the Trinidad epidemic was apparently similar to that described by Drs. Spalding and Montizambert. The disease probably originated in the southern States of North America and travelled northwards to Canada and southwards to South America, whence it was imported to this Island as already pointed out.

The main difficulties which presented themselves in the diagnosis of the disease in Trinidad will best be appreciated when the features and peculiarities of the epidemic have been considered.

Definition of the Disease.—A communicable febrile disease characterized by definite periods of incubation, invasion and eruption, the last passing through successive stages of papule, vesicle, pustule and crust.

INFLUENCE OF (1) AGE, (2) SEX, (3) RACE, (4) SEASON.

(1) *Age* (*see* Appendix, Table II.).

The youngest patient attacked was aged 2 weeks, whilst the oldest was aged 89. Adults were far more frequently affected than children; 56·20 per cent. of my cases occurred in adults between the ages of 20 and 34, whilst only 12·23 per cent. occurred amongst children aged under 14. This is exactly what one would expect in an epidemic of small-pox occurring in a vaccinated community such as exists in Trinidad, where a Vaccination Ordinance which is strictly enforced requires the successful vaccination of all infants before the age of 3 months. Again, amongst adults more were affected during the quinquenniad 20 to 24 than during any other, and amongst twenty-one children under 5 years of age, twelve were unvaccinated infants whose ages ranged from 2 weeks to 4½ months (*see* Tables III. and IV.). These figures clearly indicate the rôle which vaccination played in connection with the disease.

Even the fœtus was sometimes attacked, and the earliest period at which this occurred was after four and a half months of intra-uterine life. Four such cases came under my observation. Fifty-one pregnant women were admitted to the maternity ward after recovery from the disease. Eleven aborted and nine were delivered prematurely. In the aborted cases, eight of the fœtuses showed distinct evidence of an attack of the disease; and of the prematurely born four showed external manifestations of it, including a case of twins. All those who were attacked were born in the eruptive stage of the disease except one, which presented the characteristic macules on the body and a deep scar on the left cheek. The history of the twin case referred to above is interesting.

The mother developed the initial symptoms of small-pox on March 23 and the rash appeared on the face on March 26. Three days before the onset of the invasion period she was vaccinated, and both vaccinia and variola ran their course concurrently; the vaccine vesicles were typical and the attack of small-pox was moderately severe. On April 17, when she was in the desquamating stage of the disease, she was delivered prematurely at the seventh month of twins. Both fœtuses showed the eruption of small-pox—white macerated vesicles—sparsely scattered on the scalp, face and trunk and limbs, including the palms and soles (fig. 1). There was one large placental mass which was partially implanted in the lower uterine segment. Each fœtus was enclosed in a separate bag of membranes. The first fœtus was stillborn and the second died a few minutes after birth. In this case the fœtuses were apparently infected simultaneously, or almost simultaneously, with the mother. In the other cases there was no correspondence as regards date of disease in mother and child, although allowances were made for the peculiar conditions which affect the evolution of the rash in the fœtus. The disease was much more advanced in the mother than in the fœtus. It would appear, therefore, that either the incubation period is longer in the fœtus than in the adult or that the fœtus becomes infected after the disease has reached the eruptive stage in the mother. The liability of the fœtus to the disease seemed to decrease directly with its age.

The remaining thirty-one pregnant cases were delivered at term, and of these one woman gave birth to a child showing evidence of having passed through a complete attack of the disease. In this case the mother contracted the disease in May, 1903, and was delivered in the following July of a healthy female child with nine macules sparsely scattered on the left cheek, right lower eyelid, right arm, both forearms and buttocks (fig. 2). A similar case came under my observation in which the mother developed initial symptoms on July 29, 1903, and was admitted to the Isolation Hospital on August 6. She was discharged well on September 5, and on the 21st of the same month gave birth to a full-term infant with twenty-seven macules scattered on the face, trunk and extremities, and one mark with a depressed centre on the right cheek. The macules on the extremities were smaller than those on the face and trunk. In no instance was the eruption copious in the fœtal cases, though the majority of the mothers were severely attacked. It was also noticed that the face was not more affected than any other part of the body; this observation supports the theory that light influences the distribution of the rash in the adult.

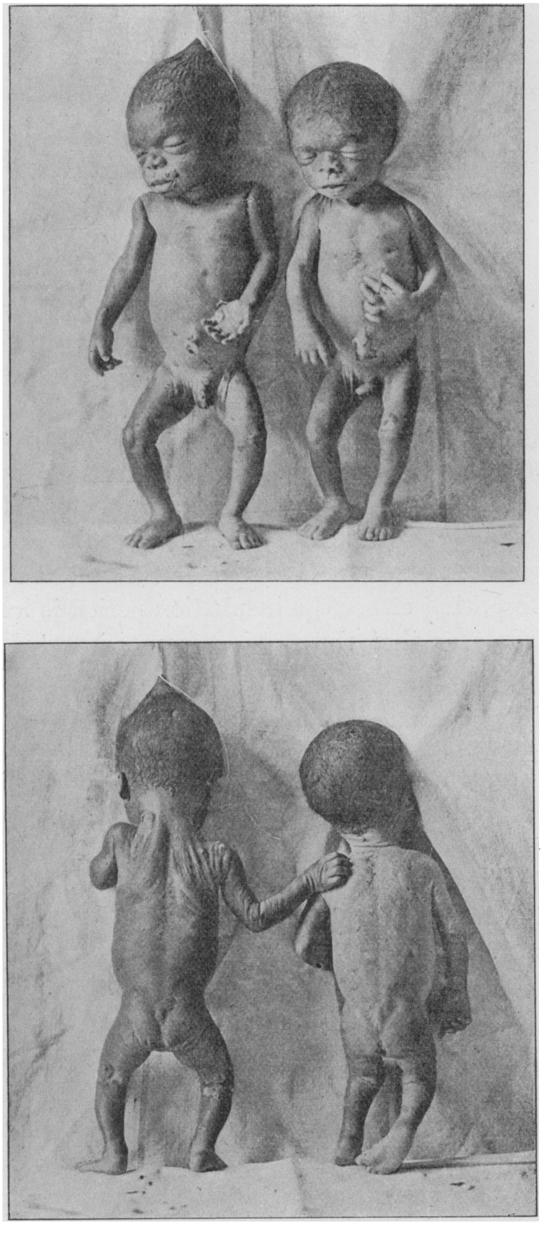


FIG. 1.

Seven months twin fœtuses, with vesicles sparsely scattered on the scalp, face, trunk and limbs, including palms of hands and soles of feet. One fœtus was born dead and the other lived only a few minutes.

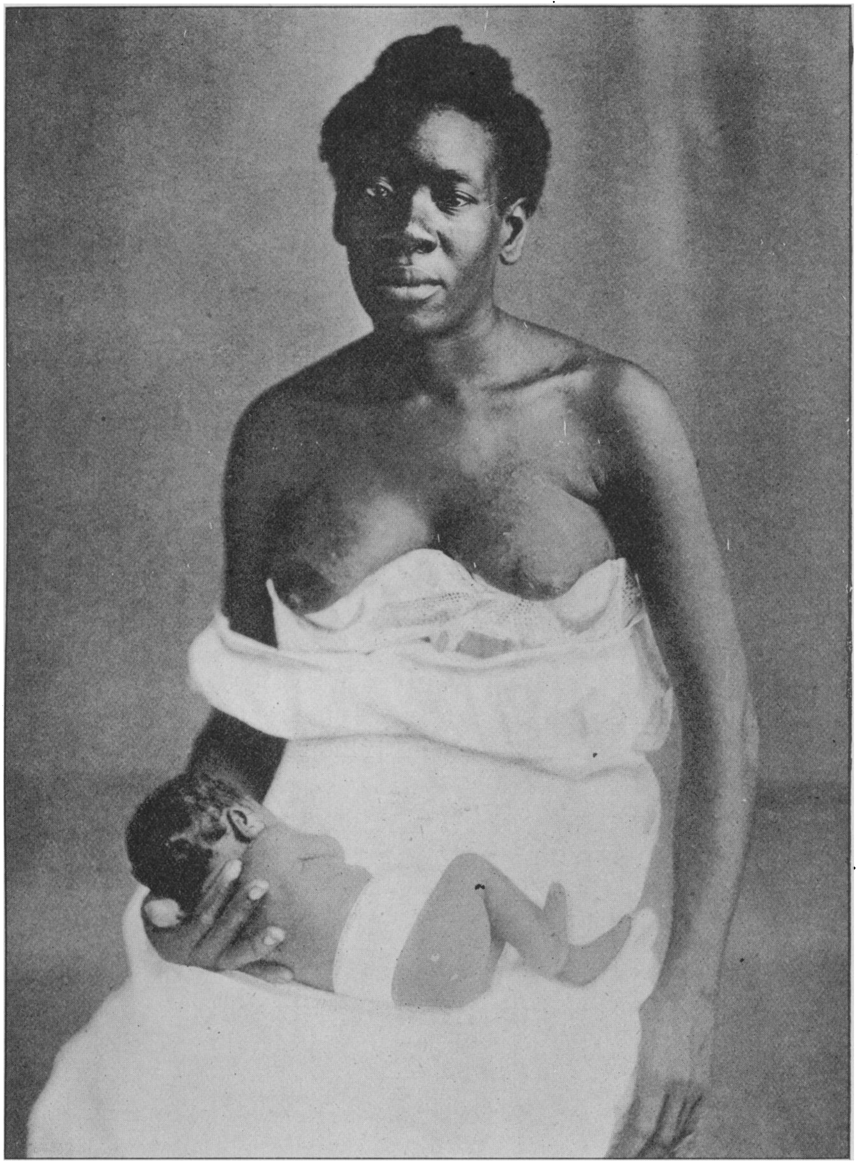


FIG. 2.

Mother, aged 28; unvaccinated. Had small-pox in May, 1903, and was delivered in following July of a full-term child. Child aged 4 days. Macules on buttock.

In the cases where the fœtus showed evidences of an attack of the disease *in utero*, we must assume the passage of the germs into the fœtal circulation. This would seem to require a breach of continuity in the walls of the maternal vessels in the placenta, if this organ acts normally as a barrier to microbes. The disease in the mother may undoubtedly produce pathological changes in this organ. Toxins and antitoxins, on the other hand, probably pass, along with nutrient matters, by osmosis. Recent investigations seem to show that the placenta has a selective power and is something more than a mere filter; in that case the existence of a placental lesion may not be necessary to explain the passage of the micro-organism of small-pox, and the transmission of its toxins may take place in a more complicated way than by osmosis.

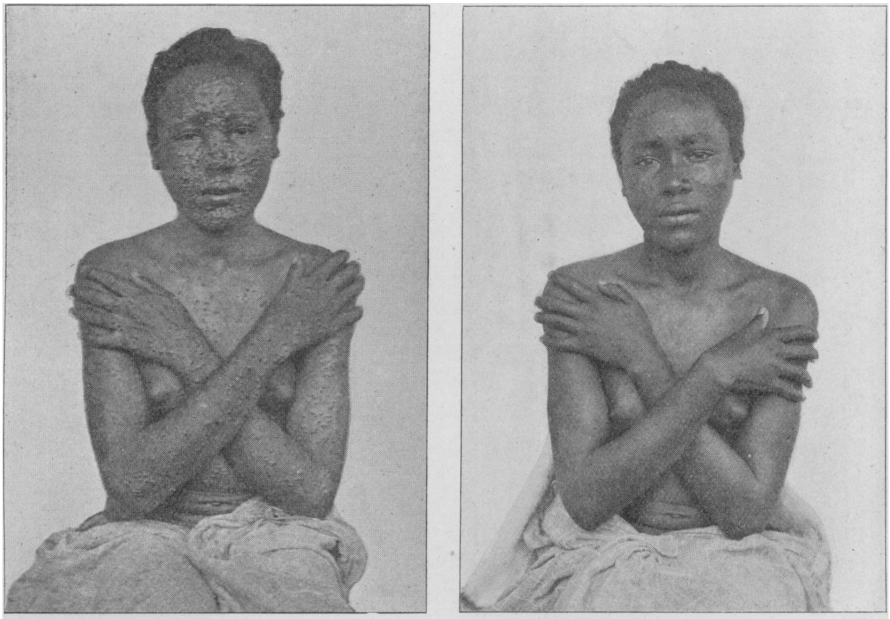
The proportion of cases of fœtal infection which came under my observation in this epidemic appears to be unusually large, but the fact that in small-pox of ordinary severity the mortality amongst pregnant women is high, and that abortions or premature labours occur more frequently, and before the external signs of the disease in the fœtus declare themselves, explains this difference.

(2) *Sex* (*see* Table II.).

Amongst my cases more males were affected than females, in the proportion of 352 to 212, and the number of attacks was greater absolutely among males than females at all ages except in the quinquenniad 10 to 14, the numbers being fifteen and nineteen respectively.

(3) *Race* (*see* Table V.).

The blacks were almost exclusively attacked, very few persons among the white section of the community suffering from the disease. This fact is in conformity with the observation that the negro race has a peculiar susceptibility to small-pox; but in this epidemic the case mortality was, contrary to all experience, exceedingly low amongst this class. A very significant fact was the immunity enjoyed by the East Indian population. This is to be attributed not to racial influence, but rather to the protection afforded by efficient vaccination and revaccination; the "coolies," as they are called here, are particularly well vaccinated. Their vaccination marks are numerous and large. There was only one East Indian amongst the 564 cases that came under my care. The estimated population of Trinidad is 280,000, and that of the East Indian section of the community is 90,000. One of the most popular suburbs of Port-of-Spain is peopled mainly by East Indians, not one of whom contracted the disease, although many cases occurred among the blacks living in their midst. It is certain the sanitary conditions under which these people live do not account for this immunity, for their habits are primitive in all matters concerning public hygiene. This consideration, however, may be of little moment, for it is a fairly well established fact that while general cleanliness and purity of water and food are useful against all diseases, the prevalence and spread of small-pox are not affected by hygienic conditions as some of the other infectious diseases are, though naturally overcrowding favours its propagation.



(*a*) FIG. 3. (*b*)

F. W., female, aged 13 ; unvaccinated.

(*a*) Photo taken on fifth day of illness. Papulo-vesicular stage. No œdema of face.
(*b*) Photo taken on ninth day of illness. Vesiculo-pustular stage. Face puffy.

(4) *Season.*

There are two distinct seasons in Trinidad, the wet and the dry. Approximately the dry season extends from January to May, and the wet from May to December; there is usually a spell of dry weather in

September or October, which lasts two or three weeks, and is commonly known as the "Indian summer."

The disease made its appearance in April, 1902, and showed no tendency to spread during the rainy season; it was only in January, that is, at the commencement of the dry season, that it began to assume epidemic form, and it continued to increase until it reached its maximum height in May, when the onset of the rains checked it. It then gradually declined until it finally disappeared—from the town in November, 1903, and from the country districts in January, 1904. The seasonal prevalence of small-pox in the tropics has long ago been observed. As far back as the middle of the eighteenth century Holwell, in speaking of the ravages of small-pox in Bengal, thus refers to the periodicity of the disease and the influence of the seasons on it:—

> Every seventh year, with scarcely any exception, the small-pox occurred in these provinces during the months of March, April and May, and sometimes prevailed until the annual returning rains about the middle of June put a stop to its fury [*see* Table VIII.].

Incubation.

The determination of the duration of this period was surrounded with some difficulty on account of the unreliability of the patients, who for the most part were ignorant, and also on account of other unavoidable sources of error; it may, however, be stated with a fair amount of accuracy that this period lasted ten to fourteen days, and this was borne out in a number of cases which afforded a decided opportunity for judging the precise time of incubation.

Invasion.

This period was hardly ever ushered in by rigors. Headache, backache, fever, and occasional vomiting or nausea appeared without any warning. Constipation was almost invariably the rule in adults, and giddiness was often complained of. This stage lasted from one to seven days, but in the majority of the cases it was of three days duration.

(1) *Headache.*—This was not a constant symptom and was not confined to any particular part of the head; it was usually general and its intensity varied very much.

(2) *Backache.*—This symptom was rarely absent. Sometimes it was very slight, but in most of the cases it was severe, and in a few instances it was described as being very violent. In pregnant women it was frequently mistaken for labour pains, so that many of the cases were

admitted to the maternity ward, where the nature of their complaint became at once apparent.

(3) *Fever.*—The fever developed usually without any preliminary chill, at least its presence was almost invariably denied; in children it was often ushered in by convulsions. The usual conditions associated with pyrexia were present, viz., general malaise, anorexia, thirst, furred tongue, quick pulse and disturbed sleep. The temperature rose rapidly,

FIG. 4.

M. R., female, aged 28; unvaccinated. Photo taken on seventh day of disease. Vaccinated on February 25. Developed initial symptoms of small-pox same evening. Three vaccine vesicles on left forearm and variolous rash general.

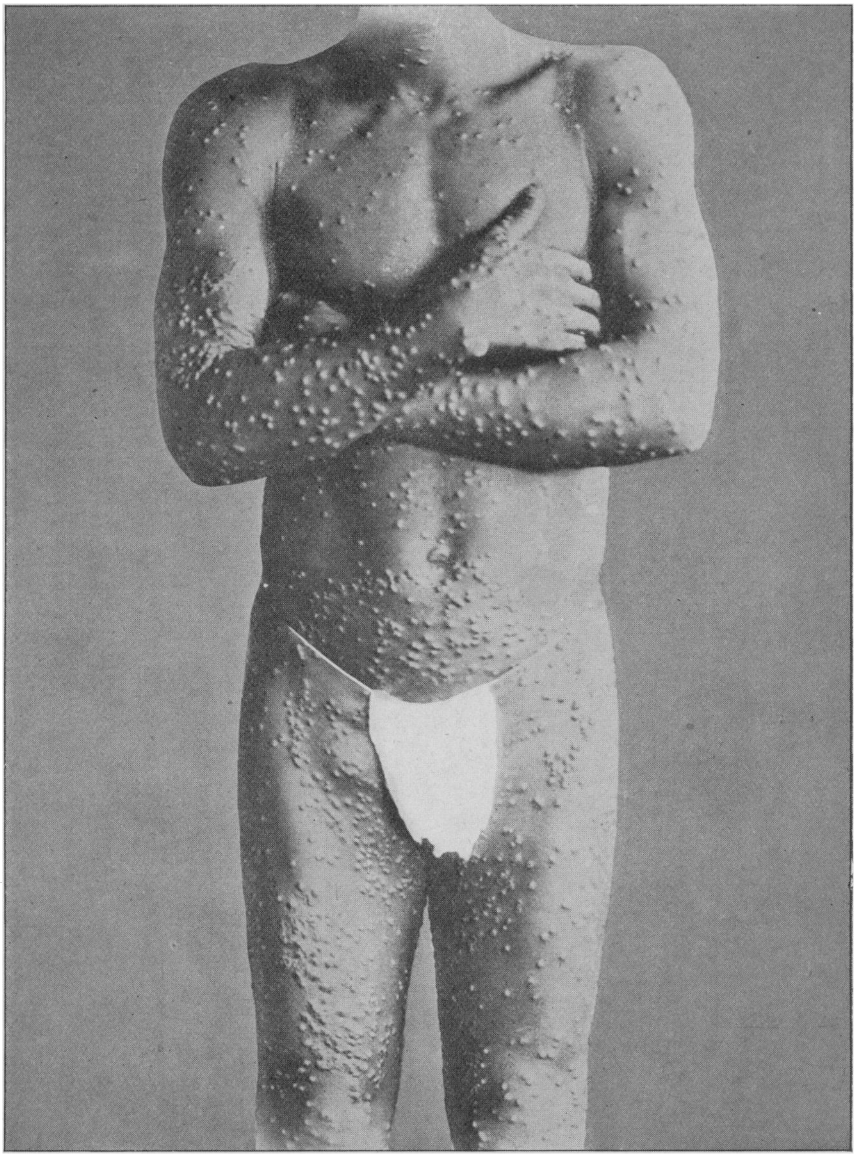


FIG. 5 (*a*).

E. S., male, aged 22; unvaccinated. Photo taken on tenth day of disease.

Front view. Large hemispherical pustules on front of thorax, abdomen, upper extremities and thighs; some of the pustules on trunk and limbs umbilicated. Gland in groin enlarged.

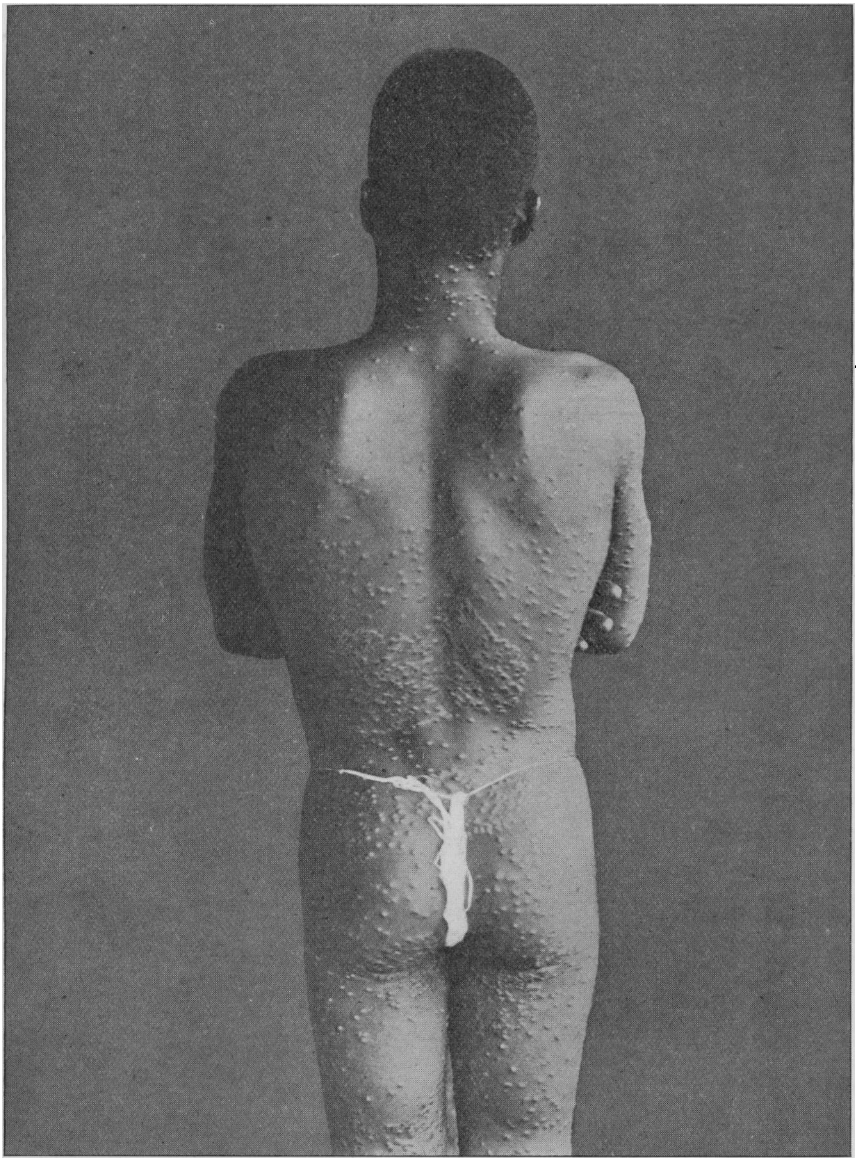


Fig. 5 (*b*).

E. S., male, aged 22; unvaccinated. Photo taken on tenth day of disease.

Back view. Shows arrangement of pustules in and around ringworm patches on the back.

and within twelve to twenty-four hours of the commencement of initial symptoms it was at its maximum height, reaching from 102° F. to 105° F. even in the abortive cases. The fever persisted with slight morning remissions as a rule during this period. Previous vaccination did not seem to influence the temperature at this stage. On the appearance of the rash it fell suddenly to normal or subnormal in the mild or abortive cases; in the severe, discrete and confluent forms defervescence was gradual, but in the latter the temperature seldom dropped to normal.

(4) *Vomiting.*—This was not a constant symptom, but it was observed in a large number of cases and was of short duration; in some instances it was, however, very distressing and persistent, causing much exhaustion. In four of my cases it ceased only when the eruptive stage was already far advanced.

(5) *Nausea.*—This occurred in a fair proportion of the cases.

(6) *Constipation* was almost invariably the rule in adults, whilst in children the opposite condition often obtained.

(7) *Vertigo.*—This was frequently complained of; most patients in this stage reeled from side to side whenever they attempted to assume the erect posture.

(8) Violent pulsation of the carotids was often observed at this period. There was no relation between the intensity of the initial symptoms and the severity of the disease, nor was there any relation between the duration of this period and the abundance of the rash. Indeed, a severe invasion period was sometimes followed by a very sparse and insignificant eruption; similarly, a long invasion period sometimes ended in a very mild attack. In infants the initial symptoms were, as a rule, so mild that the disease was often recognized only in the eruptive stage.

Initial Rashes.

No preliminary rashes occurred, as far as I could ascertain, in any of the cases that came under my care in the isolation ward. In one case a rubeoloid erythematous rash appeared on the front of the thorax of a boy during the desiccation period of the disease; it was at first very faint, then deepened in hue and gradually faded away, leaving no marks behind it.

Eruptive Period.

On the appearance of the rash all the initial symptoms of the disease subsided more or less, according to the abundance of the eruption. A sense of entire relief was experienced on the first day of the eruption in the abortive and most of the mild cases, but in the severe discrete cases this took place somewhat later, whilst in the confluent variety, owing to the painful phenomena of the eruption on the mucous membrane and of suppuration, it hardly occurred at all. The same remarks hold good as regards the temperature. In the abortive and mild cases the fever subsided at once to normal or subnormal on the appearance of the rash; in the severe discrete form this was generally accomplished only after twenty-four to seventy-two hours, owing to crops of eruption, so that there was a short intermission before the onset of the secondary fever. In the confluent cases, although defervescence took place, it did not coincide with the beginning of the eruption; it was slow and the

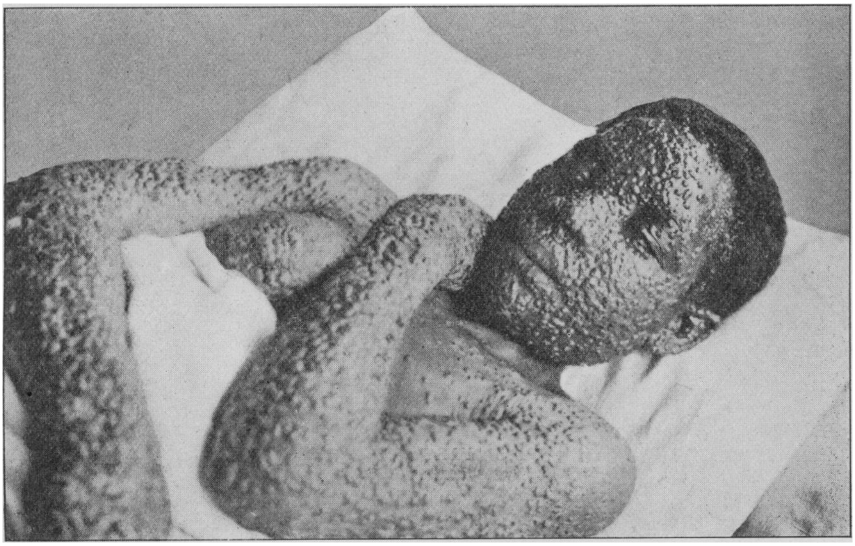


FIG. 6.

A. B., female, aged 19; unvaccinated. Photo taken on tenth and eleventh day of disease. Face puffy. Scabbing had commenced on face. Eruption was very thick on face and upper extremities, sparse on chest except on breasts, especially around the nipples. Pustules on forearms large and bullous. Did not die.

temperature rarely fell to normal, consequently there was only a remission, which was of short duration owing to the early commencement of the secondary fever. In some of the severe discrete cases there was no intermission, but only a remission of temperature; whilst in some confluent cases there was no intermission between the primary and the secondary fever, the one form merging into the other.

Generally on the fourth morning of the illness small papules appeared on the forehead and face, then on the backs of the hands and about the wrists; the eruption gradually extended to the arm, trunk and lower extremities. The rash on the face was often shotty and usually a day in advance of that on the trunk, and two or three days in advance of that on the thighs, whilst the legs and feet became affected at a still later period. During the first two or three days, especially in the severe discrete and confluent cases, fresh papules kept appearing, even on those parts which were more or less thickly covered. This probably accounts for the slow and gradual fall of temperature. These secondary papules as a rule remained small, and shrivelled up rapidly, especially in the vaccinated.

In the great majority of instances the rash first appeared on the face, the next most common site being the dorsum of the hands, a fact which I observed in 9·2 per cent. of my cases. Of eleven cases in which the rash first appeared on sites other than the face or hands, the back, forearms, thighs and buttocks each furnished two instances, and the scrotum, penis and feet one each.

The papules gradually enlarged and became hard and resistant to pressure, and in about twenty-four to thirty-six hours they were transformed into vesicles; this change was observed to take place sometimes even earlier than this. The vesicles were multilocular and their contents were expressed only with great difficulty. Those on the trunk and limbs were sometimes umbilicated (*see* for example figs. 5, 9, 10 and 11). The vesicles increased in size until about the sixth day of the disease, when they became surrounded by an inflammatory areola which appeared red or black, according as the colour of the patient's skin was white or black. The contents of the vesicles began to become turbid and the central depression to disappear at about this time. Vesicles of unequal sizes were not unfrequently seen side by side on the same parts. On the seventh or eighth day of the disease the vesicles on the face were fully converted into pustules, and this transformation gradually extended to those on the trunk and limbs. In abortive cases, whether occurring in vaccinated or unvaccinated persons, the papules shrivelled up before reaching the vesicular stage, and in the instances where the papules had become vesicles desiccation took place soon after, before pustulation had occurred. The disease ran a similar course in a few of the mild discrete cases.

The rash was generally very abundant on the face, back of hands and forearms, dorsum of feet, buttocks and thighs. Frequently the back was markedly involved (figs. 9 and 19). The front of the thorax and abdomen were often remarkably free from eruption (figs. 6, 7, 8, 10, 11 and 12), even in the severe cases; the palms of the hands and the soles of the feet were invariably affected (figs. 1, 4, 9, 10, 14, 15, 19), even in the mild cases. In a large proportion the scalp, ears, scrotum, penis and vulva were invaded, especially in the confluent and severe discrete varieties. The mucous membrane of the lips, palate, fauces, uvula, pharynx, conjunctivæ, nostrils and meatus urinarius were not unfrequently implicated in the severe cases and occasionally in the mild discrete ones.

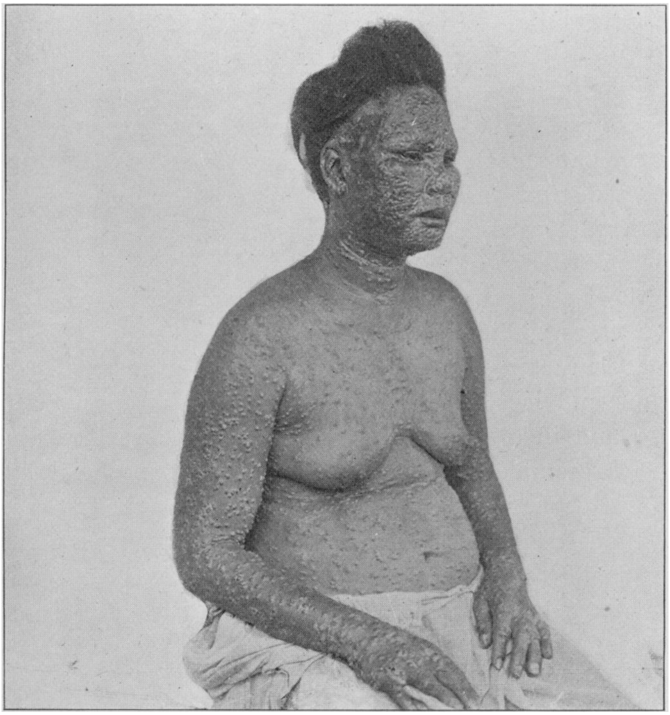


FIG. 7.

B. A., female, aged 39; unvaccinated. Photo taken on eleventh day of disease. Œdema on face had begun to subside; crusting on face. Pustules on trunk and upper arms, bursting here and there, leaving crater-like depressions in their centre—"pseudo-umbilication"; large hemispherical pustules on hands and forearms. Eruption more copious on face and upper extremities than on chest and abdomen (copious eruption on lower limbs).

Special symptoms depended on the mucous membrane affected. Thus sore throat was often complained of. The vesicles on the mucous membrane were smaller, and they matured earlier than those on the cutaneous surface; they were white in appearance; in one case, where they were not confluent, on the palate a dirty membrane was formed simulating that of diphtheria. Again, these vesicles did not maturate and scab as on the outer skin, from being constantly kept moist by secretion; for the same reason the eruption on the fœtus at birth presented a similar appearance.

The presence of the eruption on irritated surfaces was well illustrated in the case of an old man who had worn a truss for many years. The eruption followed closely the part that had been chafed by the truss, and formed a girdle round the waist. Where ringworm was coexistent with the "pox" the vesicles formed a distinct chain along the margin of the patches (fig. 5). In the case of ulcers the same ring-like arrangement was observed, and in all these situations the vesicles were larger and more advanced in development than those on other parts of the body. There was a distinct "shotty" feel of the papules, especially on the forehead, in many cases. The resisting power of the vesicles and pustules showed that they were invested with more than the mere cuticle of the skin; moreover, pitting, which resulted in a fair proportion of the cases, indicated the depth of the lesion. The bullous or pemphigoid character of the eruption on the limbs, more especially on the forearms and legs, was remarkable, and was observed in the confluent cases and in a few of the severe discrete variety (fig. 6). The bullæ closely resembled the blebs of scalds or superficial burns; their contents were dark, watery and very offensive; the temperature was septic in character. Such cases may be called "variola pemphigoides." I may here remark that the odour emitted from the cases generally was very slight, except in those referred to above. The fully developed pustules were more or less of the same size in all the cases, but there was always a variation according to the part affected in each case. The pustules on the face were invariably smaller than those on the trunk, and those on the trunk smaller than those on the limbs. The largest pustules were situated on the backs of the hands and the dorsum of the feet; these were generally about 5 mm. in diameter when fully matured (figs. 5 to 10). In a few instances the pustules were remarkably large everywhere. In the confluent form the rash on the face was small and fine, whilst large bullæ were invariably present on the limbs.

MATURATION.

This process could hardly be said to occur in all the cases, even in what seemed to be severe attacks of the disease; it began in the vesicles which had appeared first—that is, on the face—on or about the sixth day of the disease, and gradually extended to those on the trunk and limbs in the order of their appearance. The areola which had begun to form around the vesicles on about the sixth day became more extensive and inflamed on the trunk and limbs; umbilication when present began to disappear, and the pustules became hemispherical and unilocular, especially on the limbs, on about the tenth to twelfth day (figs. 5 to 19). On

FIG. 8.

P. W., female, aged 13; unvaccinated. Photo taken on eleventh day of illness. Œdema of face had not disappeared altogether; crusting on face. Pustules on arms bursting and leaving pseudo-umbilication; pustules on hands and forearms large, hemispherical and unruptured. Eruption sparse on chest.

pricking them their walls collapsed and the fluid which escaped was in some cases, even at this late stage, clear, but on pressure upon them this clear fluid was followed by sero-purulent exudation containing some solid débris. The face was generally puffy at the commencement of the maturation period, especially about the eyelids and lips, in severe cases

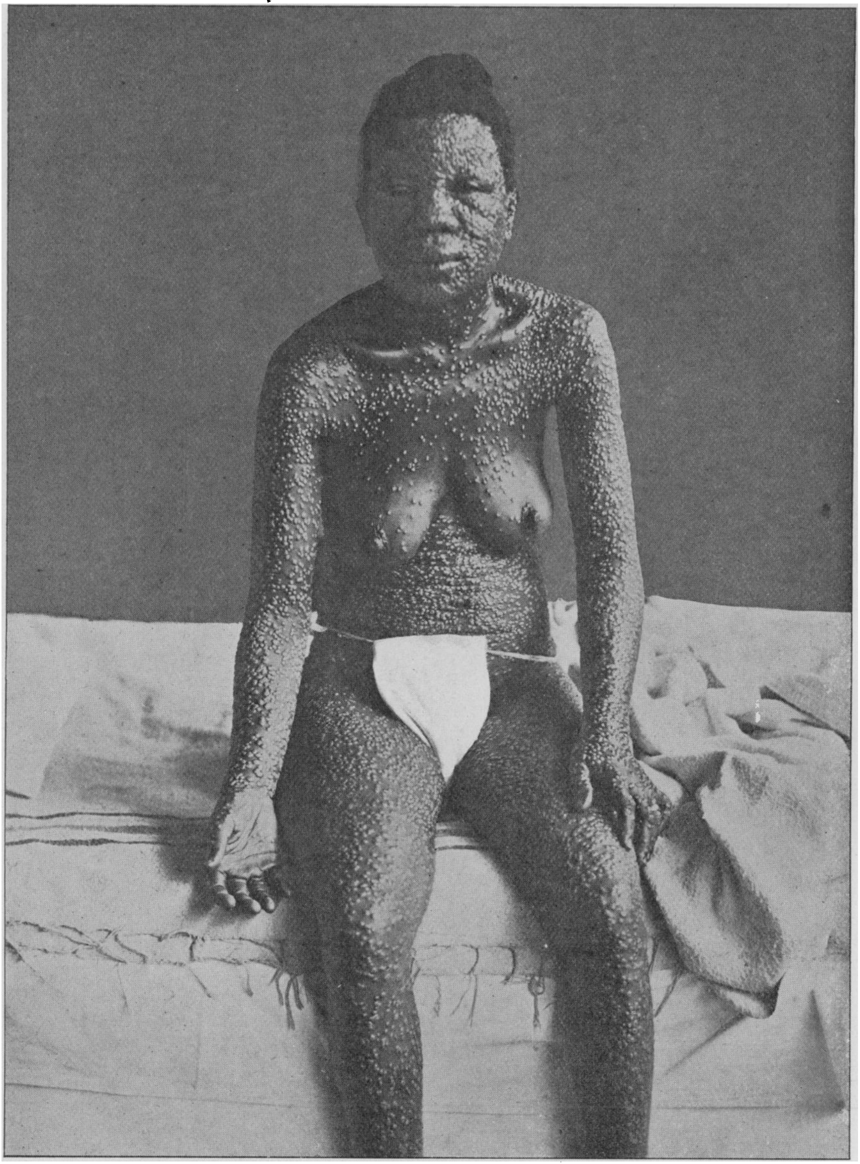


FIG. 9 (*a*).

V. B., female, aged 23; unvaccinated. Photo taken on eleventh day of disease.

Front view. Œdema of face, especially about the eyelids and lips. Eruption confluent on face. Some vesiculo-pustules on trunk and limbs, umbilicated.

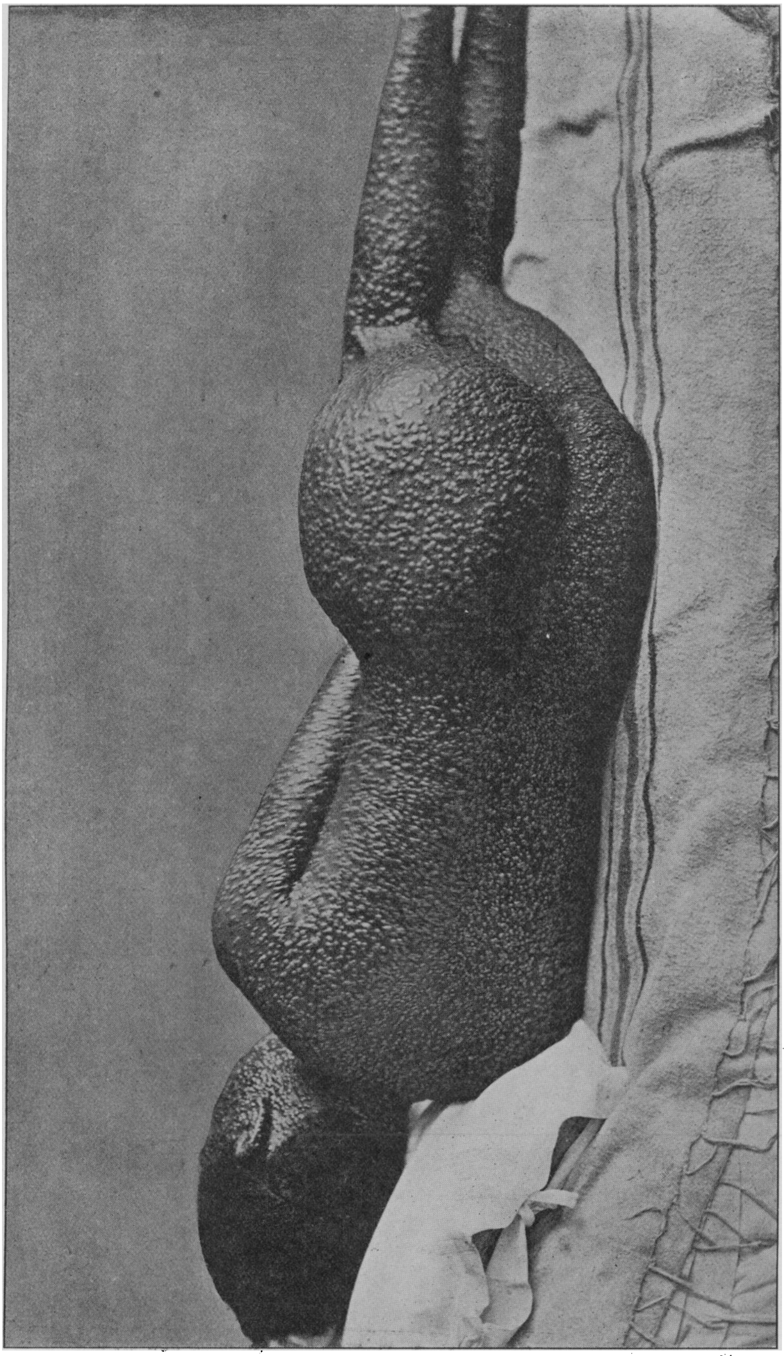


FIG. 9 (*b*).

V. B., female, aged 23; unvaccinated. Photo taken on eleventh day of disease.

Back view. Eruption very thick. Some vesiculo-pustules umbilicated; most of the pustules had lost their central depression and had become fully hemispherical. This patient was discharged well.

(figs. 3, 6 to 9, and 11). The facial œdema increased as the lesions matured, only subsiding when scabbing commenced. On about the twelfth to the fourteenth day the feet and legs sometimes became œdematous, and less frequently the hands and forearms (fig. 11). These swellings always caused considerable pain and discomfort, and so caused insomnia.

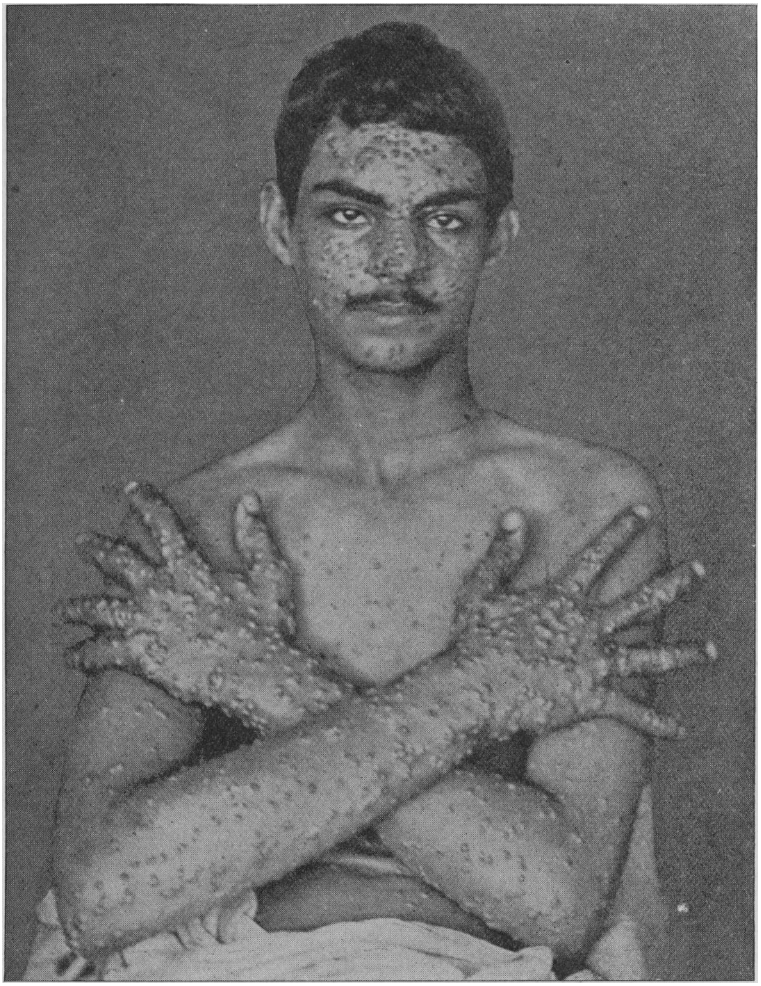


FIG. 10 (*a*).

J. de F., male, aged 20; unvaccinated. Photo taken on about eleventh day of disease. Eruption copious on face and hands, sparse on front of thorax and abdomen.

Secondary fever was absent in the abortive attacks and also sometimes in the mild discrete cases, and when present in these its intensity and duration varied very much. As a rule, there was little

FIG. 10 (*b*).

J. de F., male, aged 20; unvaccinated. Photo taken on about eleventh day of disease. Large hemispherical pustules on forearms, hands (including palms), legs and feet (including soles). Some pustules on the backs of the hands are umbilicated.

FIG. 11.

(*a*) H. F., female, aged 30; unvaccinated. Photo taken on twenty-fourth day of disease. Œdema of legs was subsiding. Pustules had burst everywhere and were drying up, leaving crater-like depressions in their centre, except a few on the insteps, which were desiccating without rupturing.

(*b*) H., female, aged 6 weeks; unvaccinated. Photo taken on fourteenth day of disease. Face and limbs thickly covered with desiccating pustules, very few on trunk. Face œdematous; crusts on face. A few pustules on insteps drying up without rupturing.

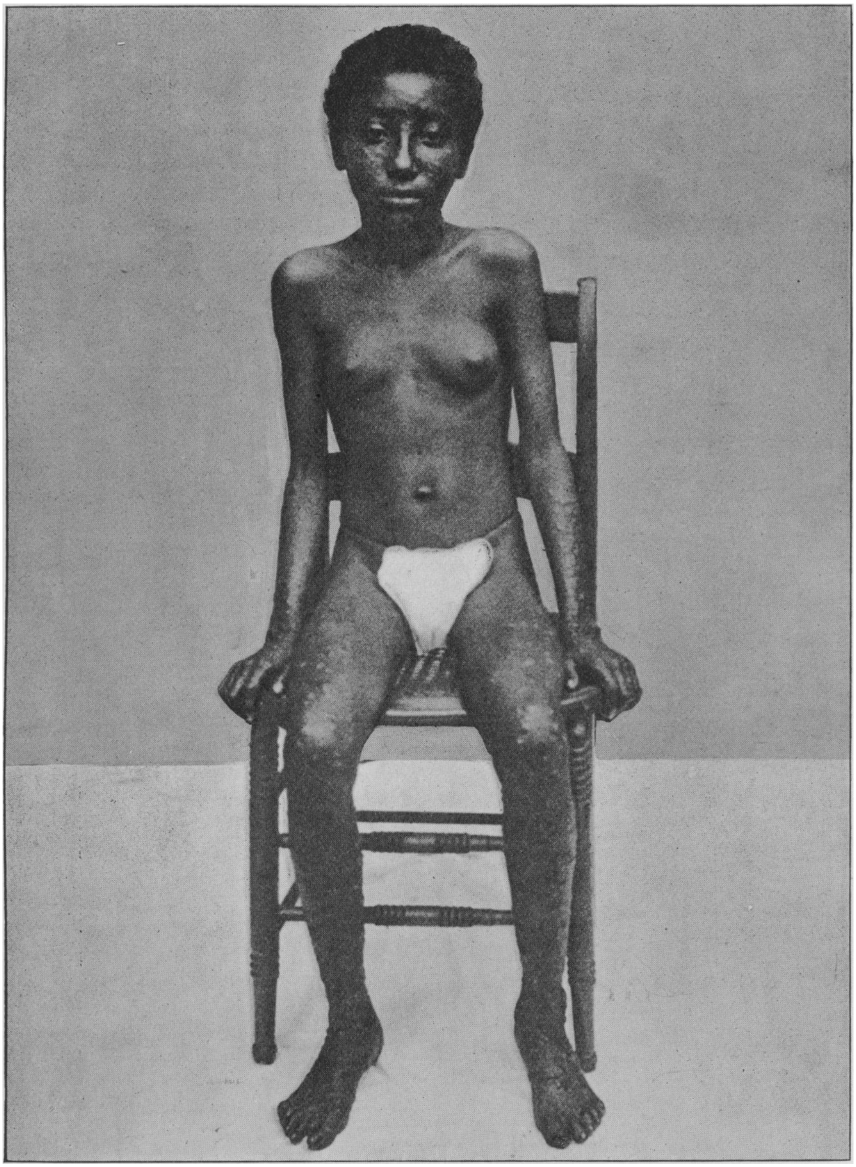


Fig. 12.

T. P., female, aged 12; unvaccinated. Photo taken on about sixteenth day of disease. Drying pustules on face and limbs, none on trunk. Scabs had nearly all fallen off face. Pustules on forearms and thighs had begun to burst; those on hands, legs, and feet had not yet ruptured.

constitutional disturbance at this period of the disease. In the severe discrete and confluent varieties the secondary fever was generally severe, but its severity was not commensurate with the abundance of the lesions. In a few instances, however, the secondary fever was very severe and prolonged. It began with the process of maturation, and its duration and severity depended more or less upon the abundance of the pustules; it lasted five or six days, but was not as high as that of the primary fever. The morning remissions were well marked. At this period of the disease, in the severe cases, all the painful and distressing symptoms of the prodromal stage returned, and to them were added pain all over

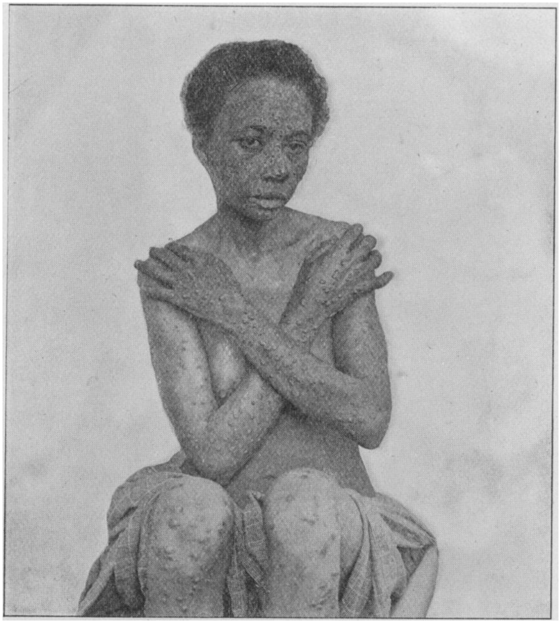


FIG. 13.

B. H., female, aged 33; unvaccinated. Photo taken on fifteenth day of disease. All œdema of face had subsided; some crusts still on face. Pustules on front of thorax and arms had burst and were drying up; those on the forearms, hands, and knees were hemispherical; some of them had ruptured.

the body due to the tumefaction of the skin, especially on the face, hands and feet, and discomfort in the throat and other mucous membranes where the vesicles appeared. Even in these cases there was, generally speaking, little depression, and the constitutional symptoms were mild in comparison with the abundance of the rash. In most of the cases the patients were able to walk about and appeared cheerful. The only inconvenience experienced by them was the pain caused by pressure on the pocks in the soles of the feet whilst walking. In a few cases, however, there was great prostration usually associated with fever of a septic nature.

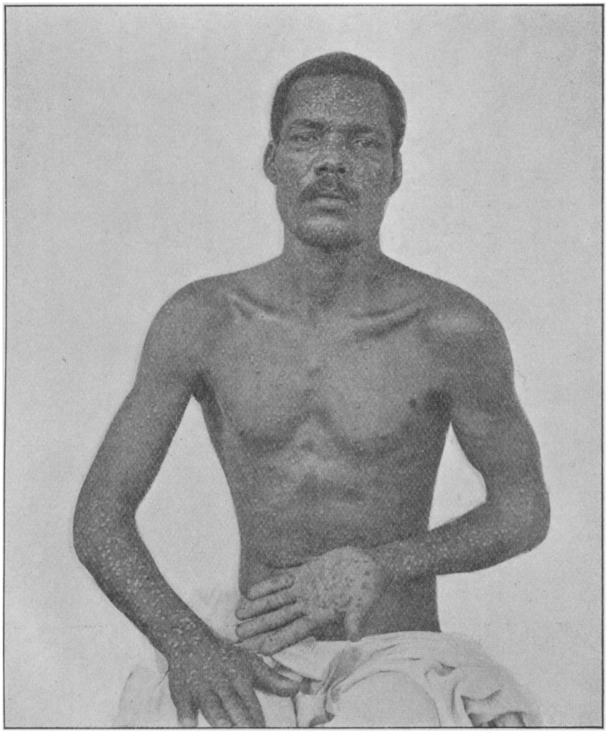


FIG. 14.

E. S., male, aged 28; unvaccinated. Photo taken on nineteenth day of disease. Warty elevations on face left after scabs had fallen off. Pustules over trunk and arms were drying up; nearly all pustules on forearms and back of hands had ruptured, those on palms unruptured; contents becoming inspissated.

Secondary fever in small-pox is generally attributed to the absorption of pus into the system from the foci of suppuration in the skin during the maturation of the pocks. If this were the sole cause of the fever, it would have been more severe and fatal, considering the extensive area of cutaneous surface involved in many of the cases that came under my care. In some instances there was hardly any healthy skin left, and yet the temperature did not rise beyond 102° F. It is also noteworthy that "cutaneous asphyxia" did not ensue in these severe cases.

Another theory regarding the cause of maturation fever assigns the pyrexia to the absorption into the blood of the decomposed discharge from the pustules. If this were so one would expect the fever to be more or less within control, but in spite of great cleanliness and the frequent use of antiseptic baths the temperature was not checked; moreover, secondary fever begins before the rupture of the pustules and terminates before they are dry. There seems, therefore, to be some

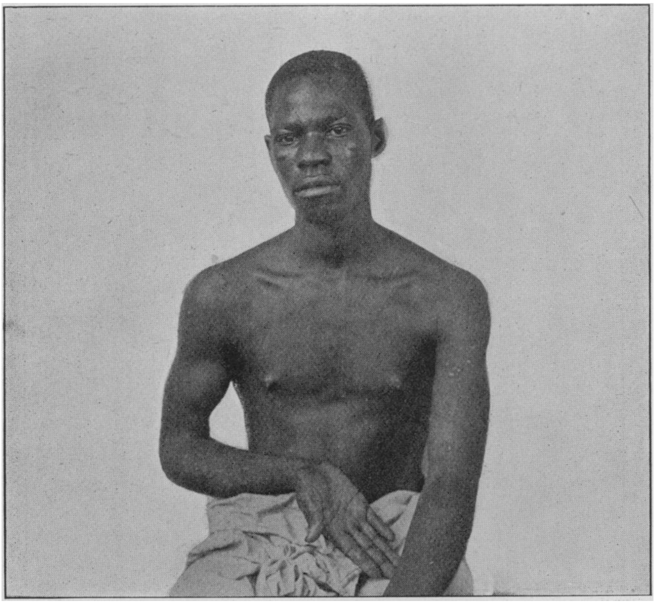


FIG. 15.

R. C., male, aged 20; unvaccinated. Photo taken thirty-one days after commencement of attack. Face pitted. Macules on front of thorax and upper extremities, including palms.

other agency at work causing the rise of temperature. It may be that a specific variolous poison is evolved in the pustules and that it produces this so-called maturation fever. The comparative mildness of the fever in this epidemic could be better explained in this way, namely, on the

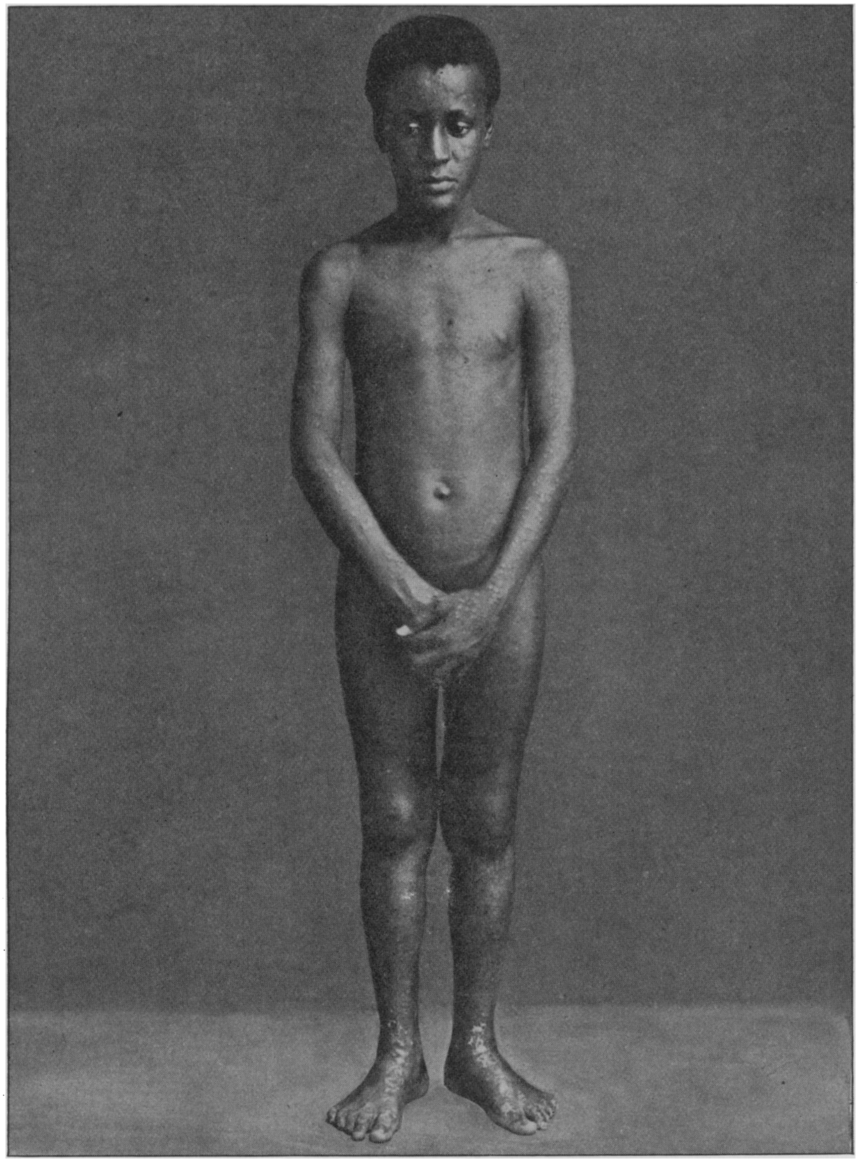


FIG. 16.

C. G., male, aged 13; unvaccinated. Photo taken forty-two days after commencement of attack. Face pitted. Macules on trunk and limbs, centre light-coloured, periphery dark.

presumption that the poison was less virulent than that which is produced in the pustules of small-pox of the ordinary type.

DESICCATION.

This process began early on the face, usually on about the eighth or ninth day of the disease; the pustules burst and the exudation from

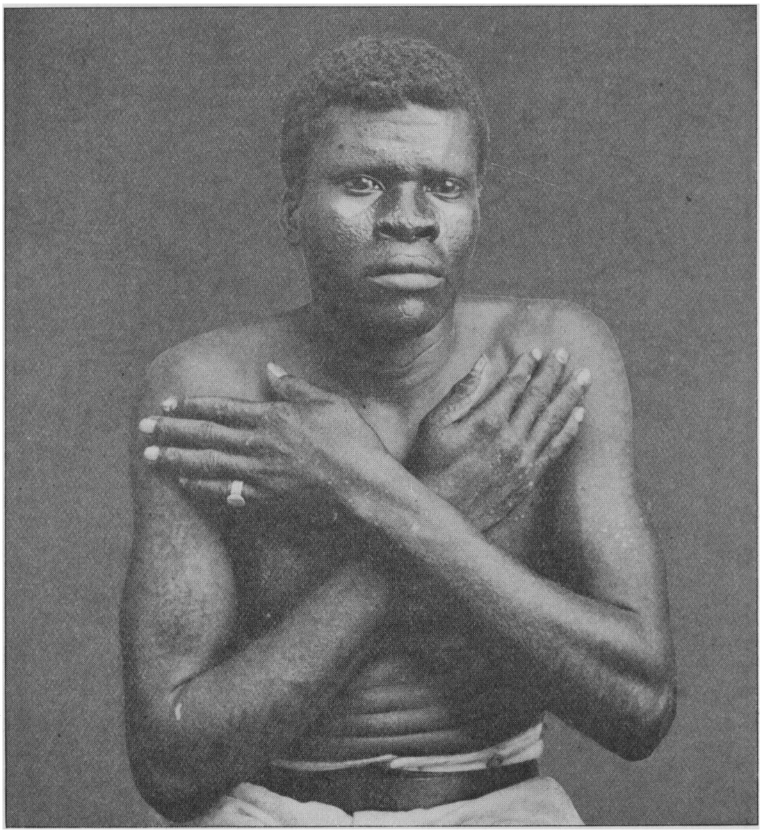


FIG. 17.

G. S., male, aged 27; unvaccinated. Photo taken sixty-four days after commencement of disease. Face pitted. Macules on trunk and upper extremities.

them caked and formed moist yellow crusts. As scabbing commenced the œdema of the face began to subside. In only two cases did the pustules on the face dry up without first bursting. When the crusts dried up and the scabs fell off on about the eleventh or twelfth day, the solid bases of the pocks remained as warty elevations on the face (fig. 14). Little by little these small pink excrescences, which were probably due to the persistent tumefaction of the papillary layer of the skin, disappeared by absorption, and in about two weeks they were replaced by macules on a level with the skin, varying in hue, but usually pink in the centre and dark at the periphery. In many cases further absorption took place until actual pitting occurred (figs. 15 to 18 and 19). This peculiar condition, so-called wart-pox, was characteristic of this epidemic and was

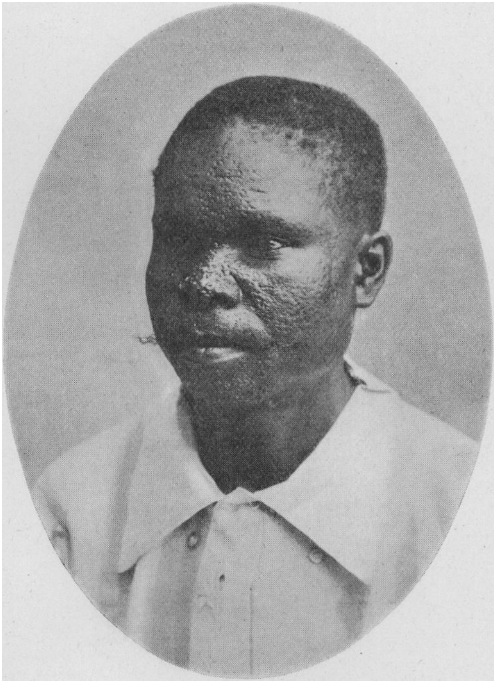
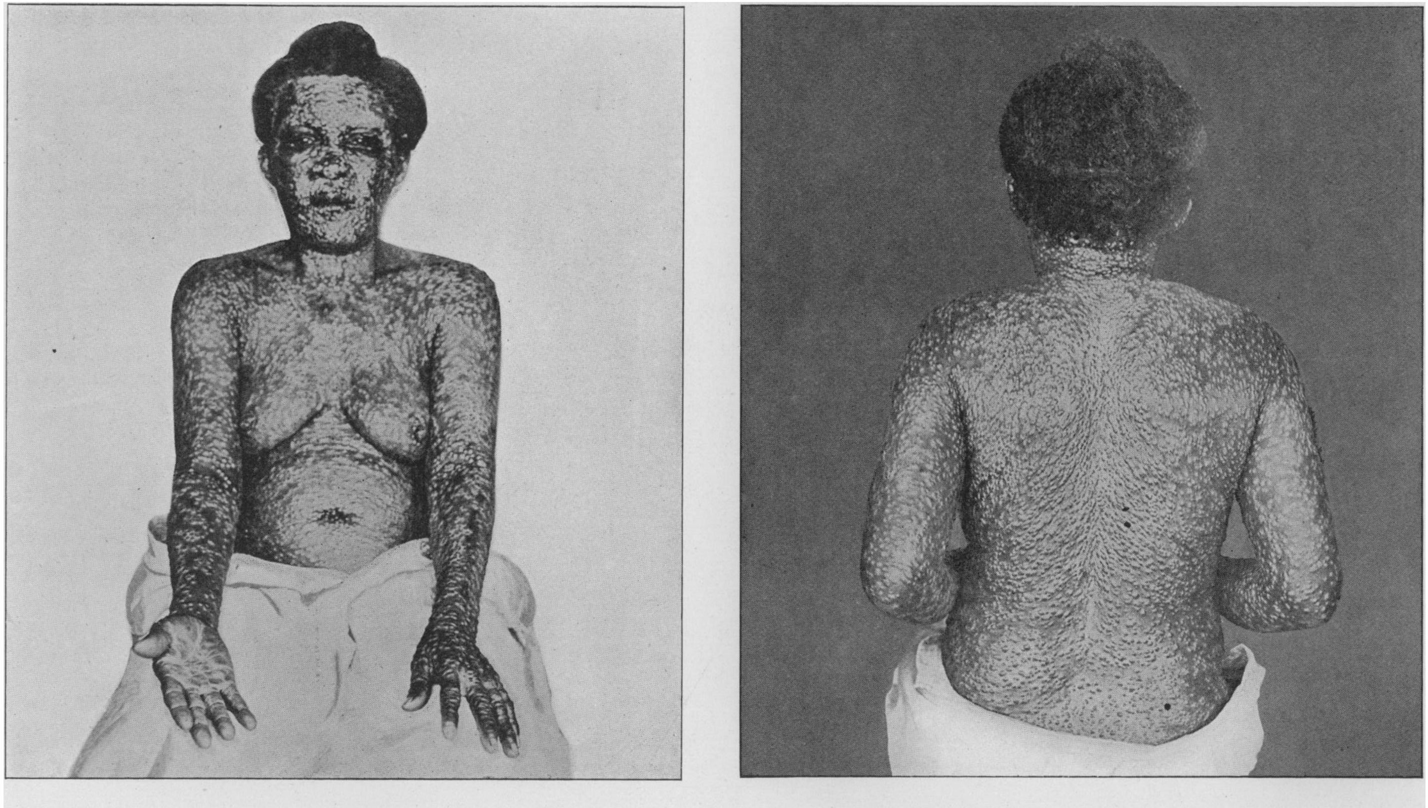


FIG. 18.

R. G., male, aged 30; unvaccinated. Photo taken seventy-four days after commencement of disease. Face pitted.

almost invariably present, even in the mild cases. It was confined to the face. In only three instances did I observe similar excrescences on the extensor surface of the forearms, and only once on the legs. Pitting occurred in one or two cases after the shedding of the scabs, notably in a recurrent case, without the process of absorption referred to above taking place.



(*a*) Fig. 19. (*b*)

M. G., female, aged 48; unvaccinated. Photo taken on about the tenth to the twelfth day of disease.

(*a*) Pustules thick on face, front of chest, and upper extremities. (*b*) Copious eruption on back. Some of the pustules still umbilicated.

About twenty-four or thirty-six hours after the commencement of the desiccation of the pustules on the face, the same process occurred on the trunk and arms, then on the forearms and thighs, and later on the legs. The pustules on the dorsum of the hands and feet were very resistant, owing to the thickened epidermis on these parts. After the rupture of those on the trunk and limbs, and the escape of their contents, small crater-like depressions were left at the bottom of the pocks (figs. 7, 8 and 11). Occasionally, the pustules on the trunk and limbs dried up

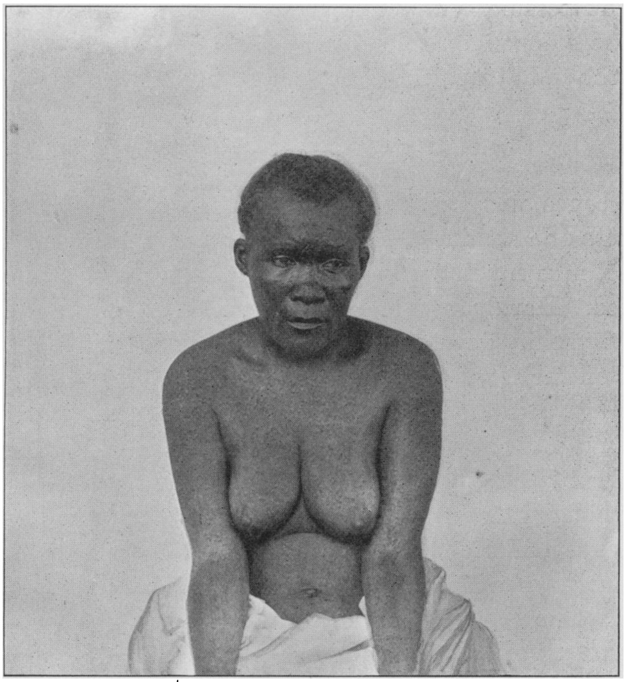


Fig. 19 (*c*).

M. G., female, aged 48; unvaccinated. Photo taken two weeks after complete recovery. Face pitted. Macules on chest and arms, centre light-coloured and periphery dark.

without rupturing and formed brownish circular crusts, which on falling off left a pink pale base gradually becoming lighter in colour and eventually fading away. The pustules on the palms and soles never ruptured spontaneously; their contents became inspissated and were absorbed; the superimposed epidermis was shed later on.

As the pustules began to burst, the areola around them faded away.

The face was the first part to clear up. The same order of succession was maintained in the desiccation of the eruption as that in which it originally appeared. Desiccation was rapid on the face, trunk and upper part of the limbs, but very slow on the forearms, hands, legs and feet, even in the mild cases. After the scabs had fallen off the trunk and limbs circular macules were left on a level with the skin, having a dark, pigmented periphery encircling a light coloured or pinkish centre (figs. 15 to 17 and 19). These marks persisted for a considerable time and then gradually disappeared after several months. During desiccation itching was often a distressing symptom. The duration of this stage varied very much, as it was dependent more or less on the severity of the attack, but in many of the mild cases the process was long and tedious on the hands and feet.

Convalescence.

Convalescence was short and rapid in the mild cases, whilst in the severe discrete and confluent forms, especially when complications arose, it was sometimes much prolonged; in the majority of the cases it was uninterrupted. Hardly any emaciation resulted except in a few instances. The patients usually developed a voracious appetite as soon as scabbing commenced on the face.

The average duration of the disease in the 564 cases I had charge of was 28·43 days; this is, however, below the mark, for unfortunately at the beginning of the epidemic many cases were discharged before they could be considered "cured," owing to the want of accommodation. The shortest cases lasted seven days; this occurred in three instances where vaccination performed during the incubation period of the disease caused its abortion. The longest case remained eighty-three days in hospital, but its detention for such a long period was due to complications and sequelæ.

Treatment.

This resolved itself mainly into questions of scrupulous cleanliness and judicious diet. In the initial stage of the disease calomel, followed by a saline purge or castor-oil, was administered; a diaphoretic mixture was then given until the rash appeared. During the eruptive stage carbolic acid or Fowler's solution was tried in several cases at first, but as no real benefit seemed to accrue from the use of these drugs they were discontinued. Arsenic, which has been lauded in the treatment of small-pox by some writers, proved positively harmful in some of my cases, as it caused much intestinal irritation which was difficult to allay. Internal medication was abandoned during the eruptive and desiccation stages except in special cases, where the use of digitalis, ether, strychnine, and other drugs was indicated. Cinchona proved very beneficial in the desquamating stage, especially where malaria was a complication. Mild preparations of iron gave good results in the anæmic cases. Opiates relieved pain and irritation, and induced sleep when most of the other hypnotics failed.

External Applications.—Tepid antiseptic baths, especially Condy's fluid properly diluted, were used from the first in almost every case, much to the comfort and relief of the patients. These baths were continued until desquamation was quite complete. Boric acid or zinc ointment and carbolized vaseline or cocoanut oil were largely employed, but the application of guaiacol in olive or cocoanut oil (1 in 80) gave the best results; it relieved itching promptly and appeared to hasten the desiccation of the pustules. This drug has been recommended by Dr. J. J. Ridge in the *British Medical Journal* of May 30, 1903.

In complications the remedies used were simply those for the disorders occurring in uncomplicated states. The wards were well ventilated, and although there was at times some overcrowding the death-rate did not seem to be affected thereby, as it invariably is in the case of small-pox of a virulent type.

Diet.—During the invasion and early eruptive stages of the disease there was, as in all febrile disorders, anorexia, and only liquid nourishment could be taken. As soon as the eruption had fully appeared the appetite was restored and a liberal diet was allowed. At the onset of the secondary fever the appetite was again impaired, necessitating a return to low diet, but as scabbing commenced on the face the patient clamoured for food. In the abortive and mild cases, where there was no secondary fever, the appetite was impaired only during the invasion period. Stimulants were given in the few instances where their use was indicated; alcohol proved very beneficial in the old and debilitated.

COMPLICATIONS.

Complications occurred in all stages of the disease and were in some instances of a grave character.

(A) In the Invasion, or Early Eruptive Stage.

(1) Respiratory system:—

(*a*) Dyspnœa occurred in five cases; in one of them it was accompanied with much pain in the chest; this condition was very distressing while it lasted, but in every case except one, where there was hæmorrhage into the lungs, it subsided on the appearance of the rash. In the other cases I was unable to detect any pulmonary or cardiac lesion to account for it.

(*b*) Hæmoptysis occurred in the hæmorrhagic case just referred to, which was the only one of this type that occurred during the epidemic.

(2) Nervous system :—

(*a*) Delirium was observed in five cases, in two of which the disease ran a mild course, whilst in the other three cases the attack was severe, and in two of them motor aphasia was also present. Delirium was more marked at night than during the day; it disappeared altogether as the eruptive stage was reached; in one case, however, it lasted ten days.

(*b*) Convulsions occurred in three female adults, two of whom had undoubtedly an hysterical tendency, whilst the third was an epileptic. In children it usually ushered in the initial symptoms of the disease and was often attributed at first to worms or dentition. This condition was of short duration and was never an alarming symptom at this stage.

(*c*) Aphasia, which sometimes occurs in acute disease, and is generally considered to be due to the toxins engendered by the specific bacilli operating upon the cells of the cerebral cortex concerned in the production of articulate speech, occurred in two cases, in both of which there was also delirium during the invasion and early eruptive stage. Both patients, after the cessation of delirium, were able to understand spoken and written speech and to translate their thoughts in writing, but the power of articulation was lost, showing that the aphasia was purely motor. Although some improvement took place during convalescence the defect of speech was still marked, even after their discharge from hospital—one forty-eight and the other forty-three days after admission.

(3) Alimentary system :—

(*a*) Diarrhœa developed in only four cases and was not a serious accompaniment at this early stage, except in the case of an infant who was already in a debilitated state.

(*b*) Melæna and hæmatemesis occurred in the hæmorrhagic case to which I have already referred.

(4) Urinary system :—

(*a*) Albuminuria was present in 18·08 per cent. of my cases in the stage of invasion, and varied in amount and duration; it vanished in the majority of the cases as soon as the rash appeared and the temperature had fallen; sometimes it persisted for a considerable period, disappearing in some instances only at the end of six weeks from the commencement of the disease. In twenty-four cases albuminuria occurred in persons suffering from chronic Bright's disease, a very common malady in this Colony, probably due to malarial infection; in these the albuminuria, of course, persisted after all the symptoms of variola had disappeared.

(*b*) Hæmaturia occurred in the hæmorrhagic case.

(5) Reproductive system :—

The catamenia frequently appeared in this stage of the disease, in many instances prematurely, but sometimes the appearance at this stage was a mere coincidence. The period was often longer and the flow more copious and bloody than normally.

(B) In the Pustular and Desiccation Stages.

(1) Respiratory system :—

Apart from a few mild cases of bronchitis and one of catarrhal pneumonia there was marked freedom from complications.

(2) Nervous system :—

(*a*) Low muttering delirium occurred sometimes in the old and debilitated, and usually signalled a fatal termination of the attack.

(*b*) Paralysis of the bladder was met with in one case at the end of the desiccation stage.

(*c*) Peripheral neuritis was occasionally observed in this stage, but this condition was more in the nature of a sequela than of a complication.

(3) Alimentary system :—

(*a*) Diarrhœa occurred in twenty-two cases at this late stage; two adults succumbed to it. In children it was a frequent complication, but no deaths resulted from it.

(*b*) Salivation was observed in only one case; it began in the eruptive stage and persisted during the maturation of the pocks; there was not much enlargement or tenderness of the salivary glands, nor was there any eruption in the mouth. I may mention that no medicine containing mercury in any form had been administered to this patient.

(*c*) Vomiting, which occurred in only one case at this period, was very persistent and difficult to check.

(4) Urinary system :—

Pyuria occurred in one case.

(5) Integumentary system :—

(*a*) Boils or small abscesses were by far the most frequent of all the complications; they occurred in 46 per cent. of the cases and developed during desquamation, usually in the axilla, or on the back, thighs, and buttocks, and kept on appearing for a considerable time in some cases. They varied in size from that of a pea to that of a walnut and caused little constitutional disturbance. In one case forty-two small abscesses formed on the back of the patient.

(*b*) Carbuncles occurred in two cases and produced severe general symptoms and also much exhaustion.

(*c*) Gangrene of the toes followed in a very anæmic and pregnant woman.

(*d*) Skin eruptions appeared in many instances at this stage, especially ecthyma, acne pustulosa, rupia, and pustular scabies.

(6) Reproductive system :—

(*a*) Orchitis occurred in thirteen cases, and was accompanied by effusion into the sac of the tunica vaginalis in six instances; the fluid was always turbid. Both testicles were affected in one case and in another an abscess formed.

(*b*) Ovaritis was diagnosed in two instances.

(7) Circulatory system :—

No complication could be assigned to this system except a case of phlebitis of the brachial vein.

(8) Locomotory system :—

Synovitis of the knee- and ankle-joints occurred in a few cases, but the effusion was never purulent and was rapidly absorbed.

(9) Lymphatic system :—

Enlargement of the inguinal glands, and more rarely of those in the neighbourhood of the elbow, was observed sometimes; the pain was usually of short duration, but the swelling persisted for a long time.

(10) Organs of sense.

(i.) Eye :—

(*a*) Conjunctivitis was a rather frequent complication, especially in the maturation stage.

(*b*) Keratitis was also present in some of the severer cases, and was sometimes very rapid in its work of destruction.

(*c*) Panophthalmitis of the eye occurred in three patients; in one of them both eyes were destroyed.

(ii.) Ear :—

(*a*) Otorrhœa was observed in two cases, but yielded quickly to treatment.

(*b*) A mastoid abscess developed in one case.

Other Complications.

Malarial fever, which so often accompanies other disorders in the tropics, was observed in a few of the cases. Typhoid fever was a complication in one case.

SEQUELÆ.

(1) Respiratory system :—

Acute pulmonary tuberculosis developed in two cases, soon after desquamation was complete, and ran a very rapid course. This disease is common here, and sometimes ends fatally in a remarkably short time.

(2) Nervous system :—

(*a*) Peripheral neuritis was not an uncommon sequela of the disease; it affected usually the extremities.

(*b*) Myelitis occurred in one case.

(*c*) Insanity. A young woman who had just recovered from the disease developed acute mania.

(3) Urinary system :—

(*a*) Chronic nephritis appears to have developed in a woman during convalescence; she had small-pox when she was five or six months pregnant. Her urine was then free from albumin; it was again examined shortly before labour and found to be loaded with albumin; this persisted for three months, after which time I lost sight of her. The persistent presence of albumin in the urine in this case points to some cause other than pregnancy; at any rate the albuminuria of pregnancy is, as a rule, temporary; it usually disappears after labour.

(4) Integumentary system:—

(*a*) Pitting showed itself in a considerable number of the severe and in a few of the mild cases; it was confined to the face and affected especially the forehead, cheeks and nose (figs. 15 to 19).

(*b*) Pigmentation. After the scabs had dropped off, macules were left with a pale pink centre and a dark pigmented periphery. These marks gradually faded away, and several months after recovery disappeared entirely (figs. 15 to 17, and 19).

(*c*) Alopecia. In two severe cases the hair dropped out completely, leaving the scalp bare during convalescence, but two or three months after recovery it grew again. This condition was observed also in an infant; in this case it was only partial, the anterior portion only of the scalp being affected.

(*d*) Shedding of the nails. In several of the severe cases the toe-nails were shed without any apparent sign of inflammation; this process was probably trophic in nature. The finger-nails were less frequently affected, and at a later period than the toe-nails. Regeneration of these epidermic appendages followed in two or three months.

(*e*) Exfoliation of the skin of the hands and feet was observed in four very severe cases. The skin of these parts was cast off entire, like a glove or a slipper.

Influence of the Disease on Pregnancy.

(A) Cases admitted to the Isolation Ward.

Thirty-eight pregnant women were admitted to the isolation ward in the invasion or early eruptive stage of the disease; twelve gave birth to, apparently, full-term healthy children at this stage. In these cases the onset of labour may possibly have been precipitated a few days by the initial fever or it may have been a mere coincidence in the regular course of pregnancy. Of the remaining twenty-six women who had not completed the full term of gestation, two gave birth to premature infants and one aborted. The further history of twenty of the remaining twenty-three, who were discharged cured of the disease, was traced.

Sixteen carried the fœtus to term, three were confined prematurely, and one aborted. The age of the fœtuses in the cases of interrupted gestation ranged from six to eight months, and the date of delivery with reference to the disease in the mother was four to twelve weeks after the commencement of prodromal symptoms.

(B) Cases admitted to the Maternity Ward.

Fifty-one women who had had the disease during pregnancy and had recovered from it were admitted to the maternity ward under my care. Thirty-one had reached the full period of gestation and were delivered of healthy children—one of the children showed evidence of having passed through the disease *in utero;* it exhibited the characteristic macules (fig. 2). Of the remaining twenty women, eleven aborted and nine were delivered prematurely.

It would appear that the disorder in the initial and early eruptive stage had little or no immediate effect upon pregnancy; gestation was usually interrupted in its course four to twelve weeks after the mother had developed the prodromal symptoms of the disease. This was due either to the death of the fœtus, caused by an attack of the disease *in utero*, or to fatty degeneration of the placenta—a condition frequently observed in these cases. In the majority of instances pregnancy ran a normal and an uninterrupted course. I may here remark that potassium chlorate was administered to a pregnant case as soon as desquamation began and was continued until delivery, when a healthy child was born. The effect of the disease on the fœtus has already been described.

VARIETIES

	Vaccinated					Unvaccinated	Grand Total
	One mark	Two marks	Three marks	Four marks	Total		
(1) Abortive	6	6	3	5	20	21	41
(2) Mild discrete	21	30	4	3	58	258	316
(3) Severe discrete ...	14	6	1	2	23	165	188
(4) Confluent	2	—	—	—	2	16	18
(5) Hæmorrhagic	—	—	—	—	—	1	1
	43	42	8	10	103	461	564

The above table shows the large proportion of mild and abortive cases which occurred both in the vaccinated and unvaccinated. The proportion of mild to severe cases among those who were treated in their homes was even greater than is shown by this table, for as a rule only the worst cases were removed to hospital. These mild cases presented well-marked irregularities not only in the initial symptoms, but also in the evolution of the eruption. The main irregularities in the symptoms were:—

(1) The occasional absence of headache or backache and, in three instances, of fever.

(2) The almost entire absence of constitutional symptoms in many instances, these patients being able to pursue their daily labours without discomfort or inconvenience.

(3) The complete absence of secondary fever, or when present its extremely short duration, as a rule lasting only a few hours.

Irregularities in the evolution of the eruption showed themselves in the abortive development of the rash; the papules often shrivelled up before being transformed into vesicles, and even when the papules became vesicles these frequently desiccated without previous pustulation ("variola vesiculosa" [Thomas] or "variola varicelloides"). As these peculiarities occurred in such a large proportion of the cases, in the vaccinated as well as the unvaccinated, the epidemic may perhaps be considered to be the mildest which has yet been recorded. The anxiety and alarm usually apprehended in the more familiar form of the disease were conspicuously absent in the community during this epidemic.

Comparison of Vaccinated and Unvaccinated Cases in respect to liability of Attack.

Vaccination had a decided influence upon the disease; of the 564 cases that came under my care, 103 occurred in vaccinated, and 461 in unvaccinated persons. The patient's word as to the success of previous vaccination was not accepted without verification by careful examination of the scars. Among the vaccinated the proportion attacked was in an inverse ratio to the number of marks present. Thus forty-three cases occurred amongst those who showed one cicatrix, whilst there were only eight cases among those with three scars. The percentage of mild or abortive cases was greater in the vaccinated than in the unvaccinated, and no confluent or hæmorrhagic case was observed in the former class.

All the deaths—thirteen in number—occurred in unvaccinated subjects (*see* Table VI.). These facts indicate clearly the rôle played by vaccination in relation to the disease.

Observations on the incidence of the disease among the nursing staff afford a striking confirmation of the previously stated facts regarding its relation to vaccination. Thirty-six nurses, eight ward-maids and three ward-men were employed in the Isolation Hospital (*see* Table VII.). *Of the thirty-six nurses only three contracted the disease, and these three had never been vaccinated;* of the remaining thirty-three, seven, who were successfully vaccinated a week to four years previous to their joining the staff, escaped the contagion; of the remaining twenty-six, fifteen were successfully revaccinated one week to four years previous to their attendance on the small-pox patients and did not contract the disease. Of the remaining eleven five were revaccinated without success and were not attacked by the disease. Three of the remaining six were revaccinated after they had been ten, fifty-two, seventy-seven days, respectively, in the isolation wards, but only one of these reacted to the operation. One nurse had been vaccinated three times without success; another was vaccinated at the age of 12 successfully and suffered from an attack of small-pox the same week; in June, 1903, she was revaccinated with negative result; another, who had never been vaccinated, contracted the disease before she joined the staff.

As regards the eight ward-maids, the only one who was never vaccinated took the disease. Of the seven others two were vaccinated, four revaccinated successfully shortly before their services were engaged, and one was vaccinated in childhood and showed a very large cicatrix on the arm. Of the three male attendants who showed doubtful vaccination marks, one was successfully revaccinated a few days after he came into the ward; five days after this he developed the disease in an abortive form, vaccinia and variola running their course concurrently. Another was vaccinated three times with success and did not contract the disease, whilst the third had already contracted the disease before he was employed.

Reference has already been made to the immunity enjoyed by the East Indian population. The evidence, therefore, of the influence of vaccination upon the disease is strong, and is in conformity with the experience of all observers.

The following cases of the disease deserve separate notice on account of the special features which each presented.

Case I.—Hæmorrhagic Case.

B. L., a well-nourished and muscular negro, aged 30, unvaccinated, began to complain of general malaise on June 8, 1903, but was able to perform his usual work on that day. On the morning of June 9 he felt worse and took to bed; he then had fever and severe pain in the back; these symptoms, with the addition of headache from the 10th, continued unabated until the 12th, when he noticed what he described as prickly heat (lichen tropicus) on the hands and feet. On the appearance of this rash the general symptoms subsided. On June 11 his eyes had become very bloodshot; on June 13 he began to pass blood in his urine and to expectorate blood-stained sputum, and on the evening of that day his motions were observed to be black and tarry. He was admitted to the Isolation Hospital on June 14 at 4 p.m. in a very critical condition; his face was puffy and covered with an erythematous blush, but no distinct eruption could be detected. There was a purpuric rash on the trunk and limbs, which was rather abundant on the back of the hands and forearms and the dorsum of the feet and also on the back. The conjunctivæ were injected and the lips swollen and bleeding. On the palate was a pseudo-diphtheritic membrane. The tongue was coated with a thick dark fur. The sputum was blood-stained and dyspnœa was urgent. The patient's mind was perfectly clear. His pulse was small and quick, and his temperature at 5 p.m. 102° F., and at 8 p.m. 100·2° F. He had no sleep during the night, and experienced very great difficulty in swallowing even liquid nourishment, owing to pain and soreness in the throat. The vesicles on the limbs were observed on the morning of the 15th to contain dark-coloured blood. The temperature at 7 a.m. was 100·2° F. He expired at 9.45 a.m., and shortly before death vomited a large quantity of bright red blood.

Post-mortem Notes.—Well-nourished, tall, muscular negro. Petechial rash on face, trunk, and limbs. Petechiæ contained blood of a dark colour. The blood generally was dark and fluid. Lungs: both bases were congested; numerous hæmorrhages into the lung tissue; the right lung was bound down by old pleuritic adhesions. Heart: slightly hypertrophied, valves healthy; no hæmorrhages into its substance or into pericardial sac. Liver: congested; hæmorrhages into its substance; weight, 5 lb. 9 oz. Spleen: very congested, not enlarged, capsule thickened, substance firm. Kidneys: large, substance pale, in a state of fatty degeneration; hæmorrhages into the pelves and calices; right kidney weighed 9 oz., left kidney 10½ oz. Bladder: contained bloody urine, but the mucous membrane was pale and normal in appearance. Stomach: contained some blood-stained fluid; hæmorrhages into its mucous membrane. Larynx: intensely congested, of a purplish hue; well-marked vesicles on base of tongue, containing blood.

Case II.—Peculiar Form of Confluent Small-pox.

G. L., aged 42, unvaccinated, was admitted to the isolation ward on March 30, 1903, with the history of having had fever, headache, and backache, followed by an eruption which appeared first on the face and hands and then spread over the body generally. The date of the first appearance of symptoms could not be ascertained with precision, but, judging from the eruption, the patient, when first seen by me, was probably in the sixth or seventh day of the disease.

Condition on Admission.—A rather weak but fairly nourished woman with a very copious, vesicular eruption on the face, trunk, and extremities. The face was covered with large flat vesicles, having a dark central depression, but ill-defined edges; there was no subcutaneous œdema, not even of the eyelids; the skin presented the appearance of coarse parchment; the vesicles on the trunk were more or less of the same character as those on the face, but were larger and bleb-like in parts; those on the legs and dorsum of the feet and backs of the hands were still larger and more bullous; there were a few vesicles on the pharynx and palate; the tongue was coated with a yellow fur; the pulse was rapid and weak; temperature, 99·4° F.; the urine contained a trace of albumin and bile; the feet and legs were swollen; the mind was clouded and the patient was somewhat restless.

Progress of Case.—Prostration increased and the mind became more confused; large bullæ formed on the trunk and extremities; some of the vesicles on these parts much resembled vaccine vesicles, and the contents were serous and bright yellow in colour; the skin generally became jaundiced. Extensive areas of epidermis exfoliated, leaving raw surfaces on the chest and limbs, such as occur in superficial burns. The skin on the buttocks sloughed away *en masse*, and the patient succumbed on April 4, 1903.

Post-mortem Notes.—Body fairly nourished and covered with a vesiculo-pustular eruption, large and flat, containing bright yellow sero-purulent fluid. Large bullæ everywhere, formed by the coalescence of adjoining pustules; the contents of these were, for the most part, serous and bright yellow in colour. The superficial layer of the skin was coming away from almost the entire surface of the body, and in some situations, especially on the buttocks, there were gangrenous ulcers. The feet were swollen. Liver: large, very soft and fatty. Spleen: slightly enlarged and soft. Gall-bladder: distended with thick, yellow bile. Heart: flabby. Kidneys: congested and fatty. This was the only case of the kind which came under my observation.

Attacks in the Recently Vaccinated.

In their standard work on the theory and practice of hygiene, Drs. Notter and Horrocks remark that "much valuable evidence has been collected of late years in regard to the duration of the protection which vaccination gives against small-pox. This evidence indicates that although the susceptibility to the operation of vaccination returns comparatively soon after primary vaccination, the susceptibility to small-pox returns but slowly, so slowly, in fact, that the power of infantile vaccination against attack by small-pox may be said to remain at least to one-half of its original extent at 20 years of age." It is interesting, therefore, to note that, among the first 4,009 cases which were reported, the Assistant Medical Officer of Health had observed the disease in twenty-eight recently vaccinated and revaccinated persons; four cases had occurred within one year of vaccination; eight within three years, and four within four years, and eleven within from four to eight years. I also observed the disease in a young married woman, aged 18, who had been vaccinated four weeks previously and who showed three good recent vaccinia scars; in this instance the initial symptoms were severe, but the rash was sparse, although every part of the body, including the mucous membrane of the pharynx and tongue, was affected; the disease ran a rapid course; most of the vesicles shrivelled up, whilst a few became pustular. The next most recently vaccinated case which contracted the disease amongst those that were treated by me was a man who had been vaccinated four years before and who presented four good vaccinia cicatrices on his arm.

All experience goes to show that the duration of the protection afforded by vaccination is limited and is directly proportionate to the number and size of the vesicles produced, but it is very remarkable that this protection was so fleeting and transient as the above cases indicate. I do not think this unusual occurrence can be explained away by assuming that the vaccine lymph was not efficacious or that the diagnosis of small-pox was faulty, for the vaccinia marks were unmistakable and the course of the disease typical of varioloid, at any rate in the case that came under my observation. Might the local manifestations of vaccinia have been produced in these instances without the absorption into the system of the immunizing substance which is supposed to be evolved in the growing vesicles ? The fact that the disease was modified, at least in my case, is a proof that a certain degree of immunity was conferred.

Does the duration of immunity depend upon the nature of the lymph, the individual, or both ? It is true that at certain seasons of the year, during the hot months, vaccine lymph suffers deterioration, but then such lymph would be inert and would produce no reaction. For the last seven or eight years glycerinated calf lymph has been used in this Colony; previous to this, vaccination was practised from arm to arm. The lymph which we now employ here is obtained from the Jenner Institute for Calf Lymph, and is kept in refrigerators until required for use. A fresh supply is received every fortnight, but the results are not always satisfactory. It would appear that the duration of the immunity afforded by vaccination depends to some extent on the potency of the lymph employed. Voigt, of Hamburg, in 1881, succeeded in inoculating a calf with human small-pox lymph and, after twenty removes in calves, used the lymph, in 1882, as a vaccine on children. In 1893, when the time came round for revaccination of the same children, the failures were more numerous than with children vaccinated in 1882 with ordinary lymph, showing greater potency of the Hamburg lymph (Edwardes). The strain of lymph therefore determines the duration of immunity. In my experience human lymph gives a greater reaction than calf lymph; the former often succeeds where the latter has failed. The vesicles are larger, and the resulting scars better marked and more persistent in those vaccinated with human lymph than in those vaccinated with calf lymph. It would appear, therefore, that the former is more potent than the latter. This may partly account for the occurrence of the disease in some of the recently vaccinated in whom glycerinated calf lymph was used. Idiosyncrasy or exceptional individual susceptibility to the contagion of the disease was probably also a factor.

SECOND ATTACKS.

The possibility of second attacks was recognized as far back as the tenth century by Rhazes, and his experience has been confirmed by many observers up to the present day. Dr. Edwardes, in his admirable and instructive book on "Small-pox and Vaccination," asks the question: "Can the same person have small-pox twice with an interval of some years between the attacks?" and answers it in the affirmative. He adds, however, that such cases—fully established—are very rare, and that the frequency of such second attacks in former times is suspicious, because measles and various kinds of false small-pox were mistaken for variola. The following extract is taken from his book:—

Dr. Kubler, a high modern authority on the subject [of second attacks] says that the once survival of small-pox afforded, perhaps, no perfect protection but a strong resistance against a fresh attack.

The German Vaccination Commission of 1884 referred to this point. Dr. Koch said that second attacks were certainly rare; in the great epidemic of 1871—2, in 12,000 cases in South Germany no second attacks occurred. Dr. Reisner pointed out that, in old times, all second attacks appear to have occurred in children, never in adults; this pointed to error in diagnosis. Dr. Grossheim, who represented the army, had only met with one peculiar case out of 22,641 in military hospitals; a man had a slight form of variola three months after the first attack. Von Kerchensteiner (for Bavaria) had never heard of an early second attack, but he believed in the occurrence of second attacks, and he himself had seen a third attack. Professor Hebra, of Vienna, had treated the patient in the first two attacks; he died in the third. Dr. Kruger had seen one certain second attack in 500 cases of small-pox observed by himself. Dr. Thierfelder had never heard or seen a second attack. Dr. von Koch had met with two in Stuttgart, both fatal; in each case the second attack was many years after the first, "a long interval of time." Dr. Siegel stated that Wunderlich found twenty-two second attacks in 1,727 cases in Leipzig in 1781; six were fatal, and one of these six patients had had small-pox already in the same epidemic.

Dr. Friedberg, cited by Lotz, reported an extraordinary case from near Breslau, during the severe epidemic of 1871—2. A child had small-pox, and the attack left several cicatrices; the child was vaccinated successfully some months afterwards, and then contracted small-pox a second time, one month after the vaccine crusts had fallen, and the second attack was fatal.

Trousseau, in his book on clinical medicine, tells of a medical student who was three times attacked by small-pox; he also alludes to the death of Louis XV. from confluent small-pox fifty years after the first attack, which occurred when that monarch was aged 14.

Dr. Savill (Warrington epidemic, 1892—3) reports a woman, aged 30, vaccinated in infancy successfully, revaccinated in 1873, that is at the age of 10, contracted small-pox probably about the same time, and yet, twenty years afterwards, had a severe attack of confluent small-pox (April, 1893), which resulted in her face being badly pitted.

In the *Lancet*, August 1, 1903, Dr. Pierce records a case of recurrent varioloid rash following vaccination. He states that a boy, aged 15, whose primary vaccination took place when aged 10 and was said to have been normal, was revaccinated on December 5, 1901, successfully; on December 24, that is, nineteen days after revaccination, small-pox showed itself, being ushered in by febrile excitement with increase of temperature, &c. The eruption developed fully and followed the normal course; the general symptoms were mild. On March 6, 1902, he developed three or four vesicles of large size about the upper lips and alæ of the nose, which were taken for herpes; the main part of the eruption appeared on March 7. A diagnosis of modified small-pox was made. In this three experienced medical men concurred. He further says that from the data available it was probable that the two attacks, whatever their nature, were identical. Even allowing that the attacks were dissimilar in character, one being, *e.g.*, varioloid and one a manifestation of vaccinia, the explanation of the attacks in view of the almost equal immunity against recurrence, mutually exhibited by the two infections, would be more or less difficult.

In Clifford Allbutt's "System of Medicine" (ii., p. 578) reference is made to this subject by Dr. T. D. Acland, in his exhaustive article on Vaccinia. He says:—

> Great variations may be met with in susceptibility to vaccinia as well as to small-pox, or any of the acute exanthems. It is commonly recognized that one attack of small-pox renders the individual more or less immune against contracting the disease again; and similarly that one successful vaccination protects, at any rate for a time, against the probability of a second successful inoculation. But it would seem that, in some persons, one attack is no safeguard against a second. This is well illustrated by a case that came under the notice of Dr. C. Allbutt, in which a woman had small-pox three times, and was also three times successfully vaccinated. Such a case seems to set at defiance all laws deduced from ordinary observations, and may be regarded as the exception which proves the rule.

It is generally recognized, therefore, that second attacks are very unusual, and their occurrence within a short period of a first attack is remarkably rare.

It is interesting to note, therefore, that from April, 1902, to May, 1903, when 4,029 cases of the disease had been reported, the assistant medical officer of health was able to record twelve cases of second attacks occurring one to seven months after complete recovery from the first attack, and running a course identical with that of the primary infection. Two such cases came under my own observation. One was treated by me in the second attack in the Isolation Hospital, and the other was seen by me in company with her medical attendant at her own house. Both cases presented unmistakable evidence of a recent (primary) attack;

they showed characteristic macules, which were scattered over the face, trunk and limbs, including the palms of the hands and soles of the feet. Furthermore, the account given by the patients themselves was in accord with the facts observed by myself. In both instances the second attack ran a mild course; the prodromal symptoms were slight and the eruption desiccated rapidly. In one case marked pitting of the face resulted after the second attack. This comparative frequency of reinfection in this epidemic was another of the many peculiarities of the "Trinidad epidemic."

It is not likely that faulty diagnoses were made. The identity of the attacks in the same epidemic could hardly have been mistaken, especially as great care was exercised in the examination of these cases so as to exclude any possibility of error, knowing the rarity of such a condition under normal circumstances.

I may here mention, however, that five cases were sent to the Isolation Hospital which proved not to be cases of small-pox. Two were cases of ordinary acne, one of malarial fever accompanied with sudamina, and two of syphilitic rashes (pustular syphiloderm). The last two are of some interest and demand special notice.

N. S., aged 21, vaccinated successfully in January, 1903, and showing four large vaccinia scars on the flexor surface of the left forearm, was admitted to the Isolation Hospital on April 9, 1903, with the history of having had fever and sore throat for three days beginning on April 5, and the appearance of a rash on the face on the second day of illness.

Condition on Admission.—A fairly well nourished man with a measly rash on the face, trunk and limbs, eyes injected, throat congested, small ulcer with dirty greyish base and angry margin situated on left side of uvula. Tongue furred. Temperature normal.

Progress of Case.—The rash, which was at first papular, became vesicular here and there on the chest and pustular on the thighs. The size of the lesions did not increase, even when the vesicular or pustular stage was reached. The ulcer in the throat rapidly grew larger, and involved almost the whole of the uvula. For a long time the measly appearance of the rash was retained, and then it became scaly everywhere except on the chest and thighs. Under large doses of potassium iodide and mercury the rash disappeared and the ulcer in the throat healed. The patient also developed keratitis, but this yielded to persistent anti-syphilitic treatment. General aches and pains were often complained of, but there was no itching in the course of the disease. Fever in this case began on April 12, and continued with irregular remissions until

April 24. I was able to trace the past history of this patient, and found out that he had been admitted on February 28 to the syphilitic ward of the general hospital "with a single indurated ulcer on the under surface of the glans penis"; a well-marked cicatrix was left to tell the tale. This case, when first sent to the Isolation Hospital, was pointed out by those who held the view that the "prevailing eruptive fever" was not small-pox, as a proof of the correctness of their opinion. This man was unsuccessfully vaccinated by me during his stay in hospital.

A. F., aged 21, vaccinated at the age of 7, showed three good stigmata on the arm; was admitted on July 3 to the Isolation Hospital with the history of having had fever and slight headache on the preceding two days, and of the appearance of a rash on the hands and feet on July 2.

Condition on Admission.—Well-nourished man, with papular rash on the back, chest and limbs. Face quite free from eruption. Some of the papules were drying up, and others were capped with a tiny drop of pus. He had a chancre on the glans penis.

Progress of Case.—The papules never increased in size, nor did they vesiculate; several acquired pustular summits; most of them became scaly and were very persistent. The patient suffered from fever until August 4; he developed iritis of the right eye on July 21, which yielded to antisyphilitic drugs. The rash disappeared entirely at the end of August. This case was successfully vaccinated by me on July 21.

The history, the existence of chancre, the presence of pyrexia throughout, the character of the eruption, which did not develop into the large full pustules characteristic of variola, the want of uniformity in the size of the lesions and their polymorphic character, the slow course of the eruption, the presence of general pains and the absence of itching all marked out these two cases from the "prevailing eruptive fever."

Results of Vaccination performed during Desquamation or soon after Recovery from the Disease.

Owing to the uncertainty which existed in the minds of the profession in regard to the nature of the epidemic, every conceivable means was adopted to arrive at a correct diagnosis, and I thought some light might be thrown on the subject by applying the vaccination test to a certain number of cases. Accordingly I undertook a series of experiments with this object in view. The results which I obtained confirmed the opinion which I already entertained concerning the variolous nature of the disease. I performed 204 primary vaccinations among adults and children who were in the desquamation stage of the disease or who had practically recovered from it. Of these thirteen did not return for inspection. Of the 191 cases that were inspected, 133 failed to react, fifty-four reacted slightly to the operation, and four seemed to be fairly successful. The "slight reaction" referred to above consisted in the delayed appearance at the site of inoculation of small red excrescences without any areola, resembling tiny mulberry growths, which dried up without further development. There was no vesiculation. In those in whom the reaction appeared "fairly successful" the vesicles were late in appearing and were ill developed; there was an absence in these cases of the inflammatory zone around the vesicles, and also a lack of general symptoms. On pricking these abortive vesicles a little viscid serum, followed by blood, exuded from them; on drying up a thin scab was formed, which on falling off left a small red excrescence, which gradually became absorbed until no trace of it was left behind. These vesicles therefore differed from the normal vaccinia vesicles in size, contents, evolution and involution. Amongst the 191 cases, four were vaccinated twice and three cases thrice, with negative results.

I also revaccinated during convalescence twenty-five cases which exhibited distinct evidence of previous vaccination. Sixteen of these gave no reaction whatever; four reacted slightly, abortive vesicles of the same character as already described being produced, and one gave a normal reaction. The others were not available for further observation. The case which gave a typical reaction was that of a child, R. H., aged 13, in whom the local manifestations were normal and attended with some constitutional disturbance. The child contracted the disease on June 15 and was vaccinated successfully on June 27, when she was practically well owing to the mildness of the attack. This child had been vaccinated in infancy and showed three good vaccination marks on the arm.

Besides these 229 cases one was vaccinated in the invasion and another in the early vesicular stage of the disease, and both gave a very slight reaction, which was much delayed. Four cases were vaccinated in the papular stage, two of which gave negative results; one reacted slightly, and the last exhibited an abortive vesicle. Thus 235 vaccinations were performed at various periods of the disease, the large majority of them being done during desquamation, with practically only one successful reaction. The vaccinations were performed in twenty-nine different series, and in fourteen of these "controls" were used, which consisted of forty-eight adults, eleven of whom had never been vaccinated before.

All the primary vaccinations were successful amongst the "controls," both as regards local and general manifestations, while there were twelve failures amongst the revaccinated. These experiments demonstrated clearly the variolous nature of the disease and the possibility of vaccinia running a normal course even after a recent attack of small-pox. It may be argued that the case of R. H. was not of a variolous nature, but I have not the shadow of a doubt in my own mind that this child passed through a mild attack of the disease.

I have not been able to obtain much information on the subject of vaccination after small-pox. Indeed, it would seem from an article in the *British Medical Journal*, January 31, 1903, p. 265, that observations on this subject are scanty and vague. The following extract is taken from that paper :—

The influence of a previous attack of small-pox on the success or failure of a subsequent vaccination is a question which has engaged the attention of several authorities. Beginning at the fountain-head, Jenner himself, in his third publication, " A Continuation of Facts and Observations relative to the Variolæ Vaccinæ, or Cow-pox," writes as follows : " Although the susceptibility of the virus of cow-pox is for the most part lost in those who have had the small-pox, yet in some constitutions it is only partially destroyed and in others it does not appear to be in the least diminished. By far the greater number on whom trials were made resisted it entirely, yet I found some on whose arms the pustule from inoculation was formed completely, but without producing the common efflorescent blush around it or any constitutional illness, whilst others have had the disease in the most perfect manner." From the figures in a table in Dr. Seaton's " Handbook on Vaccination " it appears that something like one-third of the adults who had suffered from small-pox were susceptible to the local results of vaccination in a perfect manner. In this table it is interesting to note that among the soldiers in the British Army—not recruits—the proportion of perfect success was 451 per 1,000, while among recruits the corresponding proportion was only 345. The difference suggests the element of time. The likelihood is that the interval between the attack of small-pox and subsequent vaccination was shorter on the average in the recruits than in the soldiers, the former being younger men. These statistics do not state the actual interval between the attack of small-pox and successful subsequent vaccination.

The writer of the article referred to above states that whatever may be said about exceptional susceptibility of individuals, this general conclusion is quite safely deducible from various recorded facts, viz., that local reaction of the skin, either to inoculated vaccinia or inoculated variola, does not in any way prove that the individual is susceptible to attack by small-pox in the ordinary way. The system may be protected, though the skin can still be used for the cultivation of the virus; this principle applies both to small-pox inoculation and to vaccination. In the epidemic under review the large proportion of recurrences goes to prove that even after an attack of small-pox the individual may yet be left, in some rare instances, susceptible to reinfection, and in the case of R. H. it also shows that after an attack of small-pox the individual may, though rarely, be left susceptible to vaccinia.

In the *Lancet* of October 22, 1898, Dr. Brownlee and Dr. Thomson, in an article already referred to, write as follows on the relation of vaccination and antecedent small-pox to an infectious disease which closely resembled chicken-pox and small-pox:—

> A certain amount of weight was given in the decision to the fact that three of the patients had already passed through an attack of severe small-pox, two of them comparatively recently. The small-pox in at least two of the cases was unmodified. Four of the patients were revaccinated successfully during the crusting stage, while the others had been revaccinated with success from two to four weeks before the first appearance of symptoms. It may be supposed that comparatively little value should be attached to this point, but successful vaccination of small-pox convalescents, as well as the occurrence of small-pox so soon after successful revaccination, is entirely contrary to the experience of small-pox in Glasgow. An examination into the question among the cases treated in hospital during the last five years shows that fifty-two small-pox patients were vaccinated at various periods during the stage of eruption and convalescence, some twice or even thrice, but in no single instance was any reaction manifested except in some cases a slight redness of the skin, such as might occur in the neighbourhood of any superficial wound.

The conclusion deducible from my own observations is that in rare instances vaccinia or variola may occur in a person recently attacked by small-pox and run a normal course. This subject is surrounded with difficulties, seeing that we have to deal with factors so variable as the human body and the variolous disease, and also with vaccination, performed so very differently as regards degree of efficiency.

Several cases were vaccinated in the incubation period of the disease, when they appeared to be in perfect health; the results were very interesting and showed the influence of vaccination in modifying the course of the disease when performed within a certain period after exposure to the contagion. I have a record of nine cases. Three who were vaccinated in the morning developed the initial symptoms of small-pox in the evening of the same day, and both vaccinia and variola ran their course concurrently without the one or the other being modified (fig. 4). One case was vaccinated two days before the onset of prodromal symptoms and both diseases ran a normal course In four cases the invasion symptoms of small-pox appeared at an interval of four to eight days after vaccination, and in all these vaccinia ran a typical course, but the variolous attack was modified. In one instance there was an interval of eleven days between the vaccination and the onset of initial symptoms of small-pox, and yet the latter disease was modified.

A case which at first sight appeared to be one of generalized vaccinia came under my care in the course of my observations. J. J., aged 5 months, was seen by me on October 10; he had two small ulcers on the left arm, evidently following vaccination, which the mother stated had been performed at least three weeks before in St. Vincent. The child arrived in Trinidad on October 4, and on the next day he developed fever, which was followed on October 6 by a rash which was first observed around the two ulcers on the arm. When the child came under my care there were several small ulcers around the two referred to above, and a few papules of varied sizes on the forearms, right arm, chest and abdomen. The papules became transformed into vesicles, which, on rupturing, discharged a clear, serous fluid, and subsequently ulcers were formed, and these continued to exude serous fluid for some time and then became covered with yellow crusts. I inoculated a healthy infant, with lymph obtained from the vesicles on the thigh of this child, on October 13; the result at the site of insertion was an abortive vesicle which rapidly dried up, leaving no trace behind. No general symptoms were present during its evolution and there was no areola around the vesicle. On November 11 I revaccinated the same infant with glycerinated calf lymph and two typical vaccinia vesicles developed, accompanied by the usual constitutional reaction. This showed beyond doubt that the eruption in the first child was not a genuine "vaccinide."

Vaccination of Children born of Variolous Mothers.

(*a*) Children born of mothers in the invasion or very early eruptive stage.—Two children born of mothers in the invasion stage of the disease were vaccinated soon after birth, and both "took" well. I observed that children born of mothers in this stage of the disease when exposed to the contagion contracted the disease. I saw six such cases.

(*b*) Children born of mothers in the late stage of the disease, during desquamation or convalescence.—Thirty-six children born of mothers at this stage of the disease were vaccinated within a few days of their birth; two of these showed external manifestations of having passed through an attack *in utero*. Of the thirty-six cases, twenty-five failed to react to the operation, that is, 69·45 per cent., and eleven "took," that is, 30·55 per cent. Two of the eleven successful cases did not exhibit quite typical vaccine vesicles. Of the twenty-five unsuccessful cases, eleven were revaccinated, four unsuccessfully. I vaccinated five of the remaining seven for the third time and obtained a successful reaction in two cases. I again vaccinated one of the three unsuccessful cases for the fourth time with success.

All the children were vaccinated in groups of four or five, and in every series I used controls—seventy-two in all: five adults and sixty-seven infants of the same age as the above cases. Sixty controls were successfully vaccinated, that is, 83·34 per cent., and twelve were unsuccessful, that is, 16·66 per cent. I revaccinated nine of the twelve unsuccessful cases and obtained a typical reaction in five. I again vaccinated the four refractory cases, and two of them reacted to the operation normally.

I observed that children born of variolous mothers at this late stage of the disease were not attacked, though exposed to the contagion. It would seem, therefore, that children born of variolous mothers at this stage enjoyed a certain degree of immunity, but the further history of the cases showed that this immunity was only temporary; vaccination performed at a later period proved successful in all these cases. I was also able to observe the effect of revaccination during pregnancy on the fœtus in two instances. Two infants born of mothers who had been successfully revaccinated in the later part of their pregnancy were used among my "controls," and both were refractory to vaccination. They were vaccinated a few days after birth. Three months after they reacted in a normal manner to revaccination. Two women who had the disease when one month and two months pregnant respectively gave birth to full-term healthy children who reacted normally to vaccination.

The above observations show that besides the protection afforded by contracting the disease *in utero*, the fœtus may acquire a certain degree of immunity from the mother without itself passing through a regular attack of the disease. This must take place either by the simple transmission of the already developed immunizing substances from the mother to the fœtus by way of the placenta, or (as a result of a reaction in the fœtus) to the immunizing agent passing through the same channel from the mother. By the former method, fluids which are already endowed with properties upon which immunity depends are introduced into the fœtus, whereas by the latter method these properties must first be elaborated in the fœtus before immunity is conferred. The short duration of the immunity conferred in my cases would seem to indicate that the first method was the one which was operative, the children remaining protected only as long as the immunizing substances which were transferred from the parent to them were retained.

Dr. Masson and I inoculated two monkeys with matter taken from two patients under my care in the isolation ward in July, 1903. Both monkeys were vaccinated with the same lymph, one by Dr. Masson and the other by myself. My case gave only a slight reaction, whilst Dr. Masson obtained a very successful result, which he recorded in the *British Medical Journal* of September 26, 1903, p. 779 (fig. 20).

FIG. 20.

Monkey inoculated by Dr. Masson with variolous lymph ; three vesicles were developed at the site of the inoculation.

VARIATIONS IN VIRULENCE OF EPIDEMICS AND MORTALITY-RATE.

Small-pox, like all other epidemic diseases, varies in its intensity in different outbreaks. Sydenham states that "small-pox has its peculiar kinds, which take one form during one series of years and another during another." Mild outbreaks have been observed in all ages even in pre-vaccination times, and have occasionally, we are told, been mistaken for chicken-pox.

In the great pandemic of 1871—2 this Colony suffered severely; like all pandemic extensions of the disease this was characterized by its great virulence. During that epidemic 12,531 persons were attacked and 2,449 deaths occurred. This high death-rate (19·5 per cent.) bore out the experience that in the negro and coloured races small-pox is a severe affection and attended with a high mortality.

In the recent epidemic, mildness of type was shown in the slight diffusiveness of the contagion, the insignificant symptoms exhibited by a large proportion of the cases, and in the extremely low mortality. On the other hand, a large number of vaccinated persons were attacked even more severely in some instances than the unprotected. This certainly appeared to be a very anomalous occurrence, but the term "vaccinated" cannot be considered equivalent to "protected," and the apparent anomaly may perhaps be explained in the majority of instances by the fact that in these persons the original protection afforded by vaccination had worn itself out. There were some cases, however, where vaccination was of comparatively recent date, and yet the protective power was inoperative, at any rate against attack by the disease. This epidemic maintained a degree of mildness which has never before been witnessed in this Island since the introduction of the disease by the Spaniards in the early part of the sixteenth century. Indeed, the case mortality is the lowest that has ever been recorded in any country. That only twenty-eight deaths should occur during an epidemic attacking 5,154 persons, consisting chiefly of negroes, is a result which is without parallel in the recorded history of the disease. And the fact that the disorder among infants and children was rarely fatal is also very remarkable (Tables III. and IV.). Among the 564 cases that came under my care, thirteen deaths occurred. When it is borne in mind that the worst cases were treated in the Isolation Hospital, such a result is almost incomprehensible. No small wonder that much doubt and hesitancy were felt in the diagnosis of such an anomalous form of small-pox, especially when cases literally covered all over with pocks escaped death. It would appear that a fatal termination in small-pox is not determined solely by the outward manifestations of the disease, but chiefly by the virulence of the poison which attacks the system. As the virus in this epidemic was mild, few deaths occurred, notwithstanding the abundance of the eruption in many cases.

Dr. Montizambert, in an article already referred to, says in regard to the mildness of the epidemic which visited Canada in 1900:—

> It has been suggested that the mildness of type is due to some meteorological condition. Against this theory is the fact that during the period since its commencement we have had at least one intercurrent outbreak of a very virulent form of the disease introduced from the Orient. It was quickly limited and stamped out. But in the score or so of cases that occurred the mortality ran up to over 50 per cent.

Anomalous forms of small-pox were not unknown in prevaccination times, though they were not invariably regarded as of a variolous nature. At one time an almost unanimous belief was held by the medical profession that an attack of small-pox was an absolute and lifelong protection against another attack, so that when a person who presented the traces of a previous attack became affected the disease was called "horn-pox," "water-pox," &c.

Mild epidemics of small-pox have also been described under various names. I shall refer to two classical outbreaks which occurred in Jenner's time.

At the latter end of 1789 an eruptive fever, which was known to the common people as "swine-pox," broke out in various parts of Gloucestershire and appeared to have greatly puzzled the medical men there. Jenner cut the Gordian knot by inoculating his own child, then aged about 10 months, with matter taken from its nurse, who was affected with this mysterious malady; this inoculation was successful, and the test of variolation, which was afterwards applied on several occasions, showed that the child had been protected against small-pox. From this experiment it may be safely concluded that that eruptive fever was variolous in its nature.

In the year 1807, Dr. Adams, of the London Small-pox Hospital, took matter from an outbreak of what, owing to the white appearance and small size of the vesicles, was called "pearl-pox," for inoculation purposes. The result which he obtained with this lymph was identical with that from the usual type of small-pox, showing that the disease was undoubtedly variola. Death-rates in these two epidemics were low.

The mildness of type is due either to attenuation in the virulence of the exciting cause, to a heightened resistance of the individual affected, or to a combination of these factors. We know that all organisms are susceptible of variation, especially with changed conditions of environment. Attenuated forms of bacteria are produced under injurious influences, whereas exalted virulence may be secured under favourable conditions. Most of the variations with which we are familiar are temporary, and soon disappear after a return to the normal conditions, but some become permanent and heritable even after such a return, and thus give origin to new varieties. If these variations in attenuation or exaltation of virulence can be produced by artificial means, there is no reason to suppose that spontaneous variations do not occur, especially as we know that influences capable of affecting virulence in the laboratory are operative in Nature. Indeed, we meet with varying degrees of virulence under natural conditions in the case of some pathogenic bacteria, *e.g.*, *Bacillus diphtheriæ* and pyogenic cocci. In this way may be explained, perhaps, the varying characters of epidemics.

These variations are not confined only to micro-organisms; the zoologist and botanist, by removing animals and plants to different climates and different soils, have shown that the natural forms and species are capable of alteration.

From experiments of Guillou, Thiele of Kasan, Trousseau, Delpech, and others, it would seem that there is a possibility of attenuating the virus of small-pox without the intervention of the cow.

Jenner always looked upon variola and vaccinia as modifications of the same distemper, and Somering expresses very well the identity of these two diseases thus: "Variola et vacciniæ sunt morbi, non suâ naturâ sed gradû, diversi." The most recent scientific investigations of the subject strengthen the theory enunciated by Jenner and supported by Somering. Most of those who have worked in this field claim to have obtained positive results as regards the production of typical vaccinia after one or two removes, as the result of variolation of the calf. It may be presumed, therefore, that variola and vaccinia sprang from a common stock; the former departed from the original type and, by successive reproduction in man under conditions favourable to its propagation and activity, acquired its well-known virulence. It may be that the organism of small-pox in this epidemic had degenerated or reverted to its ancestral type owing to unfavourable influences.

Predisposition is also another factor which must be considered; but it plays a less important rôle, especially in reference to small-pox; though there is a marked racial susceptibility to the disease, predisposition as applied to individuals of the same race is of minor consequence. When ordinary small-pox attacks a mixed population of whites and negroes the latter are proportionately more frequently attacked, and the attacks are more severe in this class, for the degree of susceptibility influences not only the capacity to acquire the disease, but also severity. Predisposition in mild varieties of small-pox may, however, be a more important factor than it is in the usual type of the disease. Probably to the combination of these two factors is due the mildness of type in the present epidemic.

Essentially this eruptive fever and small-pox are alike; they differ rather in degree than in kind. The absence or almost entire absence of constitutional symptoms in comparison with the abundance of the eruption; the absence of secondary fever in a large proportion of the cases; the fact that a great number of unvaccinated persons had mild or abortive attacks, whilst some of the vaccinated suffered severely; the frequency of recurrences within a short period of the first attack or after recent vaccination; the bullous character of the eruption in some severe cases; the appearance of the rash in successive crops in many instances; the apparently slight infectivity of the disorder, and its slow spread among a black population largely leavened with unvaccinated immigrants; the occasional vaccinal reaction during convalescence or after recovery from the disease, and the extremely low case mortality, especially among infants and children, are facts which are difficult to explain in association with small-pox, but in the face of other and more important and salient features which I have described, these anomalies must be regarded as of little weight as affecting the diagnosis of the disease.

When one considers the history, the age-incidence, the initial symptoms, the distribution, order of appearance, character and course of the eruption in the majority of the cases, the frequency and nature of the complications and sequelæ, the occurrence of a typical hæmorrhagic case during the epidemic, the infection of the fœtus, the influence of vaccination and other facts mentioned in this paper, I think I am warranted in coming to the conclusion and in recording the fact that the Trinidad epidemic of 1902—4 was a mild and irregular form of small-pox.

TABLE I.

Showing nationality of the 564 patients under treatment in the Isolation Hospital.

Barbados	254
Trinidad	118
St. Vincent	61
Grenada	24
Demerara	23
Tobago	21
Venezuela	19
Dominica	8
St. Kitts	8
Antigua	7
Montserrat	5
Nevis	4
Martinique	3
Cariacon	2
St. Martin	2
St. Thomas	1
St. Lucia	1
Saba	1
Colon	1
India	1
	564

TABLE II.

Showing age and sex of the 564 patients under treatment in the Isolation Hospital.

	Male	Female	Total
Under 5 years	14	7	21
5—9 ,,	7	7	14
10—14 ,,	15	19	34
15—19 ,,	43	42	85
20—24 ,,	87	55	142
25—29 ,,	79	32	111
30—34 ,,	45	19	64
35—39 ,,	27	12	39
40—44 ,,	16	10	26
45—49 ,,	8	4	12
50—54 ,,	7	2	9
55—59 ,,	2	2	4
60—64 ,,	0	0	0
65—69 ,,	0	1	1
70—74 ,,	0	0	0
75—79 ,,	1	0	1
80—84 ,,	0	0	0
85—89 ,,	1	0	1
	352	212	564

TABLE III.

Showing number of infants under treatment in the Isolation Hospital and the mortality among them.

	No. of Cases	No. of Deaths
Abortive	2	0
Mild discrete	6	1
Severe discrete	4	1
Confluent	0	0
	12	2

Remarks.—The two aborted cases occurred in infants who had been vaccinated a few days before the attack of small-pox declared itself. The two infants who died were twenty and thirty-two days old respectively.

TABLE IV.

Showing number of children aged 1—4, who were under treatment in the Isolation Hospital.

	No. of Cases	No. of Deaths
Abortive	0	... 0
Mild discrete	8	... 0
Severe discrete	1	.. 0
Confluent	0	... 0
	9	0

Remarks.—All were unvaccinated except one, aged 3, who was vaccinated at the age of 3 months and showed two good marks.

TABLE V.

Showing race of 564 patients under treatment in the Isolation Hospital.

Negroes	514
Whites	9
Mixed	40
East Indians	1
	564

TABLE VI.

Showing mortality in each variety of the disease.

	Vaccinated	Unvaccinated	Percentage
Abortive ...	0 ...	0 ...	—
Mild discrete ...	0 ...	5 ...	1·38
Severe discrete ...	0 . .	2 ...	1·04
Confluent ...	0 ...	5 ...	27·77
Hæmorrhagic ...	0 ...	1 ...	100
	0	13	

TABLE VII.

Showing incidence of the disease upon the nursing staff of the Isolation Hospital.

	Number employed	Number attacked
(1) Unvaccinated	3 ...	3
(2) Vaccinated successfully one week to four years previous to joining staff	7 ...	0
(3) Re-vaccinated successfully one week to four years previous to joining staff	15 ...	0
(4) Vaccinated successfully in infancy	8 ...	0
(5) Vaccinated thrice unsuccessfully	1 ..	0
(6) Vaccinated in infancy and contracted variola same week	1 ...	0
(7) Unvaccinated, but contracted the disease at beginning of epidemic before joining the staff	1 ...	0
	36	3

Remarks.—Of the eight who were vaccinated successfully in infancy five were revaccinated without success before joining the staff, and three after they had been ten, fifty-two, and seventy-two days respectively in the isolation ward. The last was successful.

TABLE VIII.

Total number of cases reported to week ended April 11, 1903		2,009
Total number of deaths	10	

Total number of cases reported during four weeks ended May 9 in—

(1) Port-of-Spain	1,123
(2) Country districts	96

TABLE VIII.—*(continued)*.

Total number of cases reported to week ended May 9			3,228
Total number of deaths		14	
Total number of cases reported during four weeks ended June 6, in—			
(1) Port-of-Spain	826		
(2) Country districts	147		
Total number of cases reported to week ended June 6			4,201
Total number of deaths		20	
Total number of cases reported during four weeks ended July 4 in—			
(1) Port-of-Spain	329		
(2) Country districts	123		
Total number of cases reported to week ended July 4			4,653
Total number of deaths		25	
Total number of cases reported during four weeks ended August 1 in—			
(1) Port-of-Spain	143		
(2) Country districts	53		
Total number of cases reported to week ended August 1			4,849
Total number of deaths		26	
Total number of cases reported during four weeks ended August 29 in—			
(1) Port-of-Spain	69		
(2) Country districts	68		
Total number of cases reported to week ended August 29			4,986
Total number of deaths		26	
Total number of cases reported during four weeks ended September 26 in—			
(1) Port-of-Spain	34		
(2) Country districts	56		
Total number of cases reported to week ended September 26			5,076
Total number of deaths		28	
Total number of cases reported during four weeks ended October 24 in—			
(1) Port-of-Spain	5		
(2) Country districts	32		
Total number of cases reported to week ended October 24			5,113
Total number of deaths		28	
Total number of cases reported during four weeks ended November 21 in—			
(1) Port-of-Spain	0		
(2) Country districts	24		
Total number of cases reported to week ended November 21			5,137
Total number of deaths		28	
Total number of cases reported during four weeks ended December 19 in—			
(1) Port-of-Spain	3		
(2) Country districts	9		
Total number of cases reported to week ended December 19			5,149
Total number of deaths		28	
Total number of cases reported during four weeks ended January 16 in—			
(1) Port-of-Spain	0		
(2) Country districts	5		
Total number of cases reported to week ended January 16			5,154
Total number of deaths		28	

Remarks.—From October 17 to November 21, 1903, no new cases occurred in Port-of-Spain. During the last week of November three cases were reported, and these were the last cases which occurred in Port-of-Spain in the town. No cases were reported from the country districts after January 6, 1904.